INTRODUCTION

All electrical/ electronic laboratories have electrical equipments such as power supplies, function generators, and oscilloscopes that are needed to carry out the experiments. However; some students and users do not pay an attention to the potential hazards that might be fatal if safety precautions were not fulfilled. The most common hazard of all is the electric shock.

The Electric Shock

The electric Shock is a cause of an electric current passing through the human body, and the severity depends mainly on its amount. The main source of electric shock is improper equipment grounding. Nowadays most of equipments are produced with a three wire cord and thus they are safer to use. The third wire that is connected to a metal case is also connected to the earth ground (usually a pipe or bar in the ground) through the wall plug outlet.

Basic Safety Precautions

All students and users are advised to comprehend to the following safety precautions whenever they work in the laboratories:

1. Acquaint yourself with the location of the following safety items (***Fire Extinguisher, First aid kit, Circuit Breaker, and emergency Telephone numbers***).
2. Make sure that the lab instruments are at ground potential by using the ground terminal supplied on the instrument. Never handle wet, damped, or ungrounded electrical equipment.
3. Never touch electrical equipment while standing on a damp or metal floor.
4. In an emergency all power in the laboratory can be switched off by depressing the large red button on the main breaker panel.
5. Do not wear bracelets, rings and watches during lab sessions.
6. Power must be switched off whenever an experiment or project is being assembled, disassembled, or modified. Discharge any high voltage points to grounds with a well insulated jumper. Remember that capacitors can store dangerous quantities of energy.
7. Make measurements with well insulated probes, and avoid using both hands.
8. If a person comes in contact with a high voltage, immediately shut off the power. Do not attempt to remove a person in contact with a high voltage unless you are insulated from him/her. If the victim is not breathing, apply CPR immediately continuing until he/she is revived, and have someone dial emergency numbers for assistance.
9. Use CO_2 to put off an electrical fire or a dry type fire extinguisher. Locate extinguishers and read operating instructions before an emergency occurs.

Equipments and Materials

DC Power Supply

A Dual DC power supply figure 1 can be used (1) as a voltage Source, and (2) as a current Source provided a prior setting of the unit. The DC power supply provided in the laboratory is set to operate as voltage source, and the controls are:

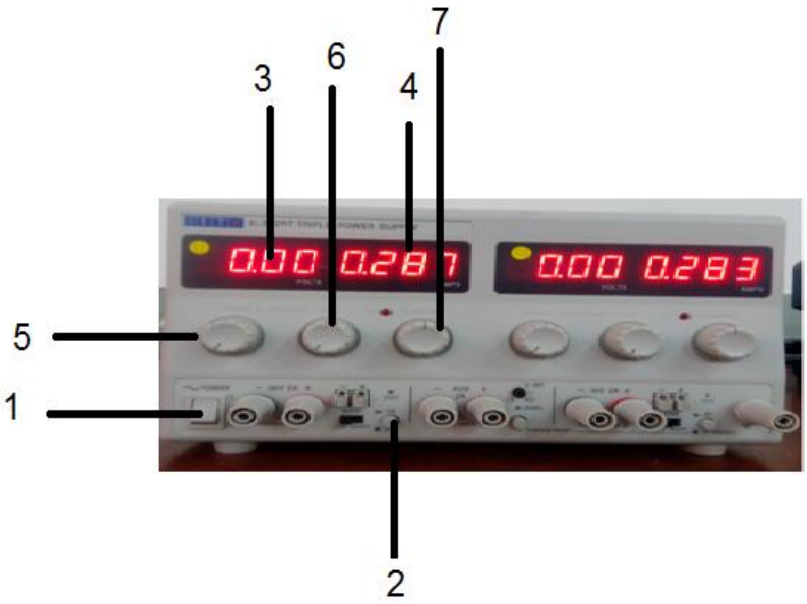


Figure 1: Digital Dual DC Power Supply

Control No.	Description
1	**Main power Switch:** Provides power to the unit
2	**Secondary Switch:** Provides power to the circuit.
3 & 4	**Voltage and Current monitors**
5	**Coarse:** Enables voltages from (0 to 30)V range.
6	**Fine:** Enables tuning the voltage value.
7	**Current Limit(CC):** Should be at Mid point to enable each source to deliver a sufficient current to the Load.

Function Generator

A function generator figure (2) is a device that can produce various patterns of voltage at a variety of frequencies and amplitudes.



Figure 2: Function Generator

The Oscilloscope

The oscilloscope, either digital or analog, is the most useful instrument available for testing circuits because it allows you to see the signals at different points in the circuit. The place or type of control buttons found on any oscilloscope front panel may differ according to the manufacturer's preference see figure (3), but the functions are the same. It draws a V/t graph (voltage verses time), voltage on the vertical or (Y-axis), and time on the horizontal or (X-axis).



Figure 3: The Oscilloscope

BNC Probe

It is a special connector that has two Ends. BNC is connected into the oscilloscope and two clip jacks that represent signal carrier and GND. The channel and the probe can be cheched by the BNC to the input of CH1 or CH2 and adjusting V/DIV to 2V/DIV. A square wave similar to that of figure 4 will be shown on the Oscilloscope screen.

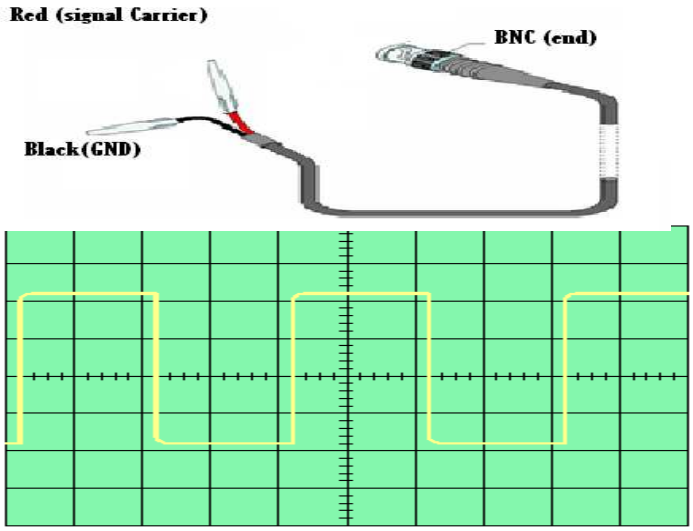
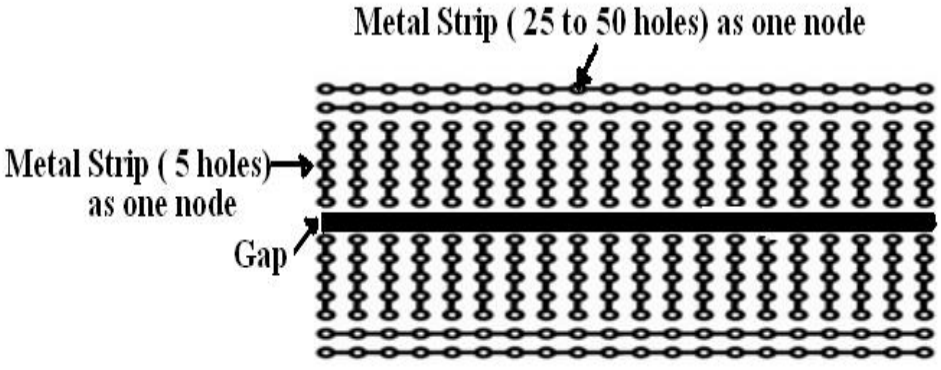


Figure 4: BNC and BNC checking

The Breadboard

It is an experimental board that has many strips of metal (usually copper) which run underneath as in figure (A). The strips are covered by a ceramic or a plastic surface that has holes that match the strips of metal and the finished breadboard is as shown in figure (B). Figure (B) also shows how the components should be placed on the breadboard.



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Figure A : connection layout diagram

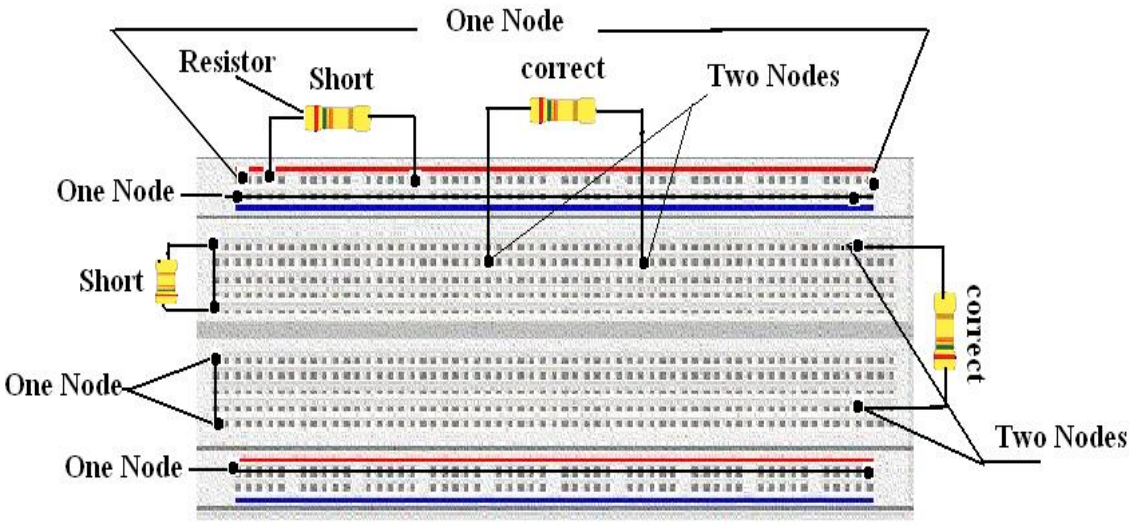


Figure B: sample connection

Electrical Components and their Identifications

RESISTORS

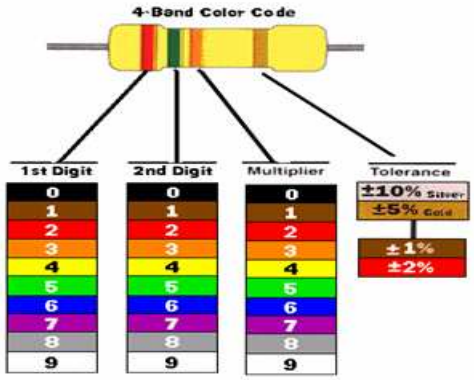
1- Fixed Carbon Resistors

A resistor value is determined according to a standard color code as shown in figure (5). In practice however; every resistor has a tolerance that determines its validity; the tolerance is determined by gold, silver or no color band.

The standard resistor color code table:

Color	Digit 1	Digit 2	Multiplier	Tolerance
Black	0	0	$\times 10^0$	
Brown	1	1	$\times 10^1$	±1%
Red	2	2	$\times 10^2$	±2%
Orange	3	3	$\times 10^3$	
Yellow	4	4	$\times 10^4$	
Green	5	5	$\times 10^5$	
Blue	6	6	$\times 10^6$	
Violet	7	7	$\times 10^7$	
Gray	8	8	$\times 10^8$	
White	9	9	$\times 10^9$	
Gold			×0.1	±5%
Silver			×0.01	±10%
None				±20%

A: color Code



B: Resistor Bands

Figure 5: Resistor identification (A): Color Code, (B): Resistor Bands

2- Potentiometer

It is a variable resistor that has three terminals as shown in figure (6). It could be used as a fixed resistor when terminals (1 and 2) are used, or as variable if terminals (1, and 3 or 2, and 3) are used. It can also be used as voltage divider when all three terminals are used provided that extreme terminals; i.e. 1, and 2 are connected to VCC, and GND. The output is taken at terminal 3.

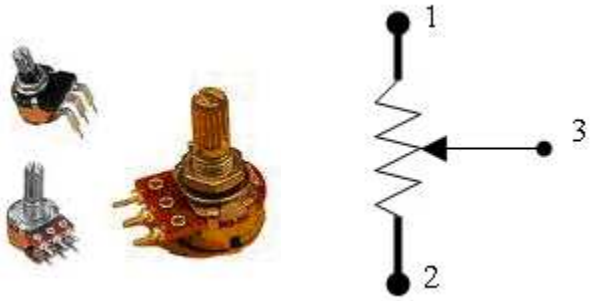


Figure 6: Variable Resistor

CAPACITORS

Capacitors store electric charge, and therefore they are used with resistors in timing circuits or filters. In amplifiers, capacitors allow AC to bypass and block DC. There are many types of capacitor, but they can be categorized into two groups, **polarized** and **un-polarized**. Each group has its own circuit symbol.

A. Polarized capacitors (large values, 1µF +)

Electrolytic capacitors are polarized, and **they must be connected correctly**. There are two designs of electrolytic capacitors; (1) **axial** where the leads are attached to each end as figure 7.b, and (2) **radial** where both leads are at the same end figure 7.c. All capacitor are subjected to voltage ratings.

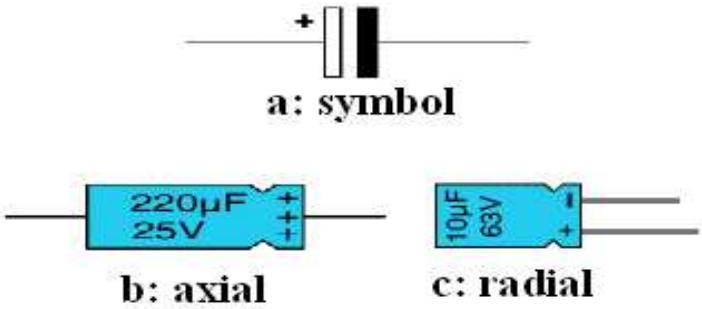


Figure 7: Polarized Capacitors

B. Un-polarized Capacitors (small values, up to 1µF)

Un-polarized capacitors have small capacitive value, and high voltage ratings (usually 50V-250V) or so. They may be connected in either way. However; determining their values are difficult because there are many types of them and several different labeling systems! Many small value capacitors have their value printed but without a multiplier, for example **((0.1 means 0.1µF = 100nF, or the multiplier is used in place of the decimal point as (4n7 means 4.7nF)).** Figure 8 shows un-polarized capacitors.

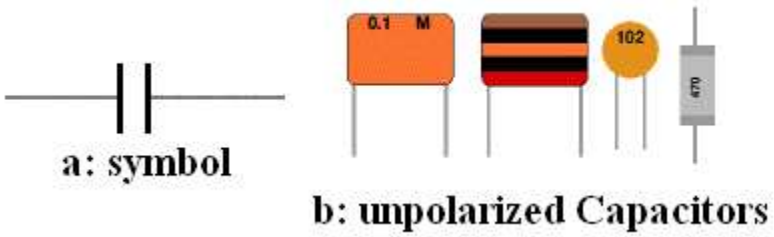


Figure 8: Un-Polarized Capacitors

To determine the value of small un-polarized capacitor, use the following code:

1) The **First** number is the **1st** digit,
2) The **Second** number is the **2nd** digit,
3) The **Third** number is the number of zeros to give the capacitance in **pF**.
4) Ignore any letters - they just indicate tolerance and voltage rating

Example

102 has been printed on a small value capacitor, determine its value in F?

1st digit = 1
2nd digit = 0
3rd is multiplier = $\mathbf{10^2}$
Then:

$$\mathbf{C = 10 * 10^2 = 1000pF}$$

Diodes Types

A diode is a two-terminal device, having two active electrodes, between which it allows the transfer of current in one direction only. Diodes are known for their unidirectional current property, wherein, the electric current is allowed to flow in one direction. Basically, diodes are used for the purpose of rectifying waveforms, and can be used within power supplies or within radio detectors. They can also be used in circuits where 'one way' effect of diode is required. Most diodes are made from semiconductors such as silicon, however, germanium is also used sometimes. Diodes transmit electric currents in one direction, however, the manner in which they do so can vary. Several types of diodes are available for use in electronics design. Some of the different types are:

Light Emitting Diode (LED)

It is one of the most popular type of diodes and when this diode permits the transfer of electric current between the electrodes, light is produced. In most of the diodes, the light (infrared) cannot be seen as they are at frequencies that do not permit visibility. When the diode is switched on or forward biased, the electrons recombine with the holes and release energy in the form of light (electroluminescence). The color of light depends on the energy gap of the semiconductor.

Avalanche Diode

This type of diode operates in the reverse bias, and used avalanche effect for its operation. The avalanche breakdown takes place across the entire PN junction, when the voltage drop is constant and is independent of current. Generally, the avalanche diode is used for photo-detection, wherein high levels of sensitivity can be obtained by the avalanche process.

Laser Diode

This type of diode is different from the LED type, as it produces coherent light. These diodes find their application in DVD and CD drives, laser pointers, etc. Laser diodes are more expensive than LEDs. However, they are cheaper than other forms of laser generators. Moreover, these laser diodes have limited life.

Schottky Diodes

These diodes feature lower forward voltage drop as compared to the ordinary silicon PN junction diodes. The voltage drop may be somewhere between 0.15 and 0.4 volts at low currents, as compared to the 0.6 volts for a silicon diode. In order to achieve this performance, these diodes are constructed differently from normal diodes, with metal to semiconductor contact. Schottky diodes are used in RF applications, rectifier applications and clamping diodes.

Zener diode

This type of diode provides a stable reference voltage, thus is a very useful type and is used in vast quantities. The diode runs in reverse bias, and breaks down on the arrival of a certain voltage. A stable voltage is produced, if the current through the resistor is limited. In power supplies, these diodes are widely used to provide a reference voltage.

Photodiode

Photodiodes are used to detect light and feature wide, transparent junctions. Generally, these diodes operate in reverse bias, wherein even small amounts of current flow, resulting from the light, can be detected with ease. Photodiodes can also be used to generate electricity, used as solar cells and even in photometry.

Varicap Diode or Varactor Diode

This type of diode feature a reverse bias placed upon it, which varies the width of the depletion layer as per the voltage placed across the diode. This diode acts as a capacitor and capacitor plates are formed by the extent of conduction regions and the depletion region as the insulating dielectric. By altering the bias on the diode, the width of the depletion region changes, thereby varying the capacitance.

Rectifier Diode

These diodes are used to rectify alternating power inputs in power supplies. They can rectify current levels that range from an amp upwards. If low voltage drops are required, then Schottky diodes can be used, however, generally these diodes are PN junction diodes.

Small signal or Small current diode: These diodes assumes that the operating point is not affected because the signal is small.

Large signal diodes : The operating point in these diodes get affected as the signal is large.

Transient voltage suppression diodes - This diode is used to protect the electronics that are sensitive against voltage spikes.

Point contact diodes

The construction of this diode is simpler and are used in analog applications and as a detector in radio receivers. This diode is built of n – type semiconductor and few conducting metals placed to be in contact with the semiconductor. Some metals move from towards the semiconductor to form small region of p- tpye semiconductor near the contact.

Avalanche diode

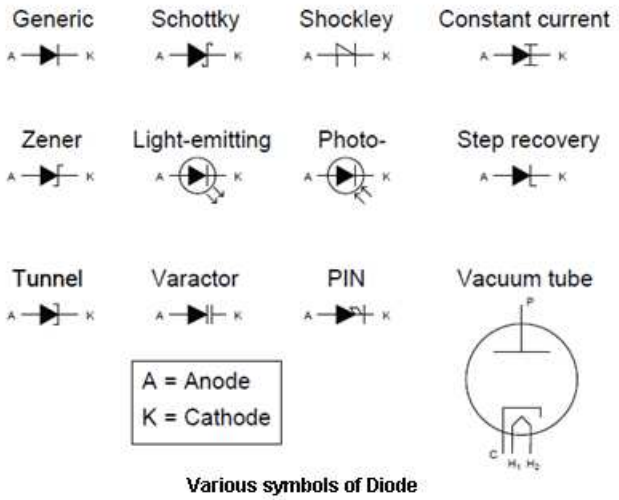
This diode conducts in reverse bias condition where the reverse bias volage applied across the p-n junction creates a wave of ionization leading to the flow of large current. These diodes are designed to breakdown at specific reverse voltage in order to avoid any damage.

Silicon controlled rectifier

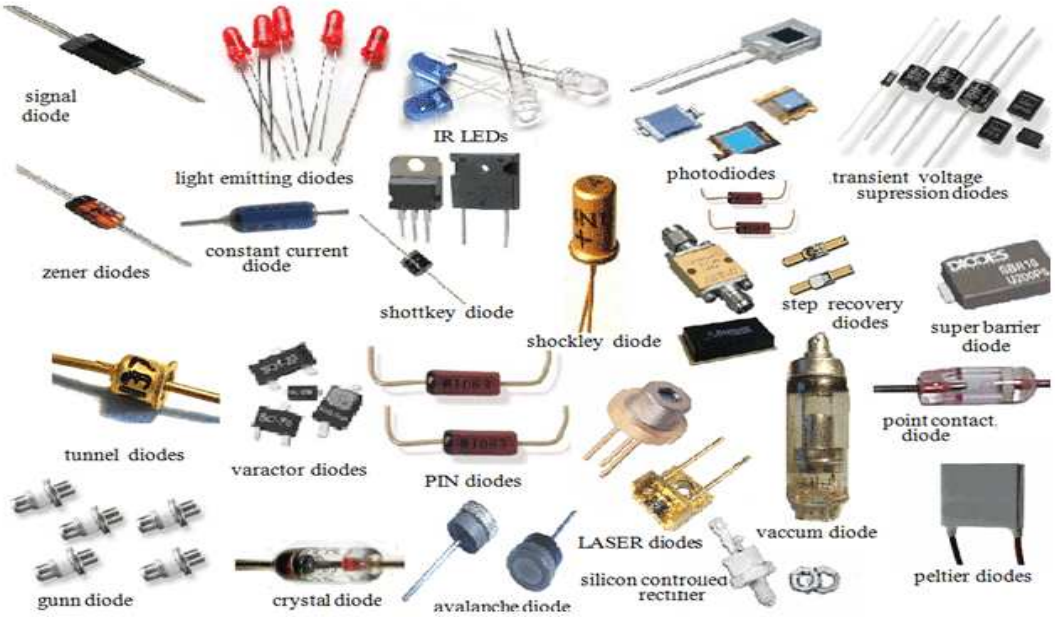
As the name implies this diode can be controlled or triggered to the ON condition due to the application of small voltage. They belong to the family of Tyristors and is used in various fields of DC motor control, generator field regulation, lighting system control and variable frequency drive . This is three terminal device with anode, cathode and third controlled lead or gate.

Vacuum diodes: This diode is two electrode vacuum tube which can tolerate high inverse voltages.

Diodes are used widely in the electronics industry, right from electronics design to production, to repair. Besides the above mentioned types of diodes, the other diodes are PIN diode, point contact diode, signal diode, step recovery diode, tunnel diode and gold doped diodes. The type of diode to transfer electric current depends on the type and amount of transmission, as well as on specific applications.



Various symbols of Diode



Types of Diode

The Technical Report

The manual aims helping the students to assemble, analyze and then write well documented reports. The students are advised to read and comprehend to the following guide lines:

1. General Guidelines

- ☒ Each experiment has been written in a structured logical sequence that will lead you to a specific set of conclusions. Be sure to read the experiment procedure carefully. Refer back to text if theory in the manual is not sufficient for you to analyze.
- ☒ Pre-lab means that you use the useful formulas to calculate expected values. This must be done before the lab sessions.
- ☒ When making measurements, check for their sensibility.
- ☒ Record your observations and draw you graphs to scale.
- ☒ All graphs must be labeled and explained briefly.
- ☒ The report should be concluded by a conclusion based on the calculations and the observations.
- ☒ **It's unethical to "fiddle" or alter your results to make them appear exactly consistent with theoretical calculations.**

2. Lab Reports

The following format should be adhered to by the students in all their laboratory reports:

☒ **Objective**

This should state clearly the objective of the experiment. It may be the verification of law, a theory or the observation of particular phenomena. Writing out the objective of the experiment is important to the student as it emphasizes the purpose for which the experiment is conducted.

☒ **Brief Theory**

The related theory of the experiment must be discussed **briefly**. It assists the student in making a conclusion based on comparison between the experimental results to the theory.

☒ **Results**

All experimental results which have been approved by the lab instructor (including graphs) must be attached in the report.

☒ **Discussion & Conclusion**

The student must form some elaboration on the results of his analysis. Usually this involves deducing whether the final results show that the aim of the experiment has been achieved or not.

EXPERIMENT ONE

GENERAL PURPOSE DIODE

Objectives

- ☒ To study the Characteristics of Silicon PN Junction Diode .
- ☒ To determine static and dynamic resistances in forward biased graphically.

THEORY AND BACKGROUND

The diode is a two-terminal semiconductor device that allows current to flow in only one direction. It is manufactured from silicon or germanium and its characteristic curve is nonlinear as shown in figure 1.1. Depending on the type of the applied voltage, its resistance at any point on the curve can be determined. For example, if a dc voltage is applied, then the type of resistance is static which satisfies equation (1).

$$R_S = \frac{V_D}{I_D} \qquad (1)$$

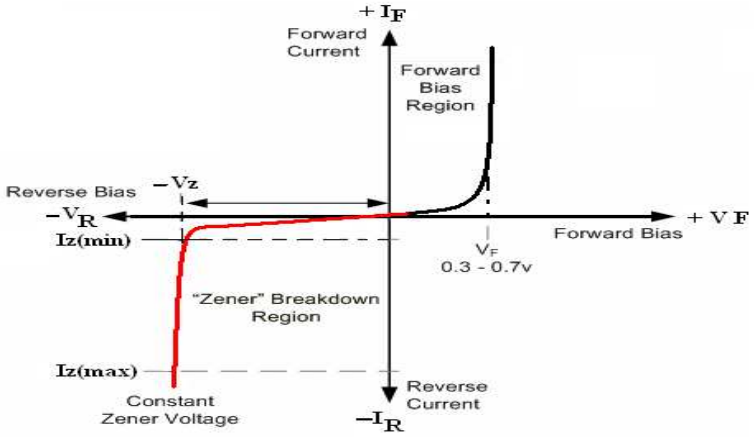


Figure 1.1: Diode I-V Characteristics

However; if an Ac voltage rather than a DC voltage is applied as in figure 1.2A, the resistance is dynamic. This situation defines a specific change in current and voltage as shown in figure (1.2B) and is given by:

$$r_d = \Delta V / \Delta I \qquad (2)$$

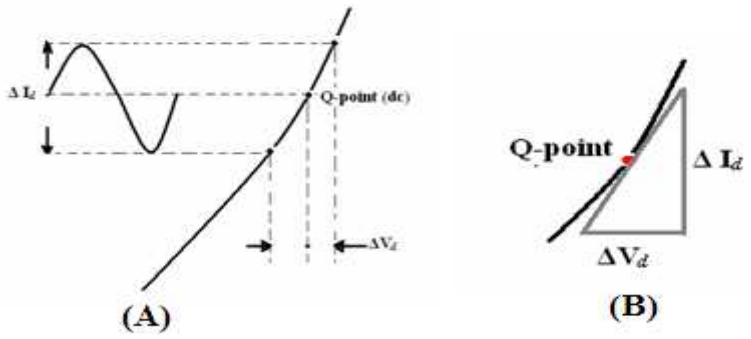


Figure 1.2: dynamic or ac resistance.

Tools and Equipments Required

- ☒ **DC Power Supply**
- ☒ **DMM (Digital Multi Meter)**
- ☒ **IN 4001 Diode.**
- **1 k Ω Resistor.**
- **Breadboard.**

PROCEDURE

1- Select Diode Symbol (►) on the DMM and test the given diode.
2- Wire the circuit of figure 1.3.

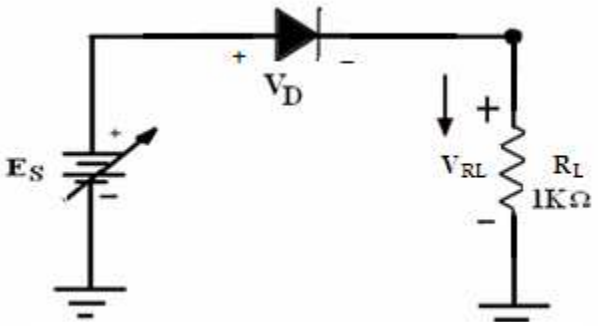


Figure 1.3: Dc diode Characteristics

1. According to table 1.1, Set (E_S) to the voltage between (0 – 3)v by Increasing the dc supply voltage in small steps around 0.2V for each step while simultaneously measuring the voltage across and the current through the diode. In the vicinity of the knee voltage (approximately 0.5V), make these steps approximately 0.05V. Record your data in Table 1.1.
2. Now, turn off the power and reverse the diode. Again set (E_S) to voltages between (0 – 5)V. measure V_R and I_R and record your results in table 1.1.

Table 1.1: Part One Data

E_S (V)	Forward Bias				Reversed Bias	
	Calculated		Measured			
	V_F (V)	I_F(mA)	V_F (V)	I_F(mA)	V_R (V)	I_R(μA)

3- Use a proper scale, Plot only V_D versus I_D on graph paper, and then graphically draw the load line and determine the static resistance R_S at (E_S = 1.5V) .

4- If R_L is made 2.2K Ω, what is the new Q-point?

EXPERIMENT TWO

RECTIFIER CIRCUITS

Objectives

- ☒ To demonstrate the operation of unfiltered and filtered rectifier circuits.

Tools and Equipments Required

- ☒ **12Vrms/50 Hz C.T. Transformer**
- ☒ **4 IN 4001 Diodes**
- ☒ **4.7 µF, 100µF, 220µF Capacitors.**
- ☒ **Dual Trace Oscilloscope**
- ☒ **DMM (Digital Multi Meter)**
- ☒ **1kΩ ,(2* 220Ω), 100Ω resistors**
- ☒ **Breadboard**

THEORY AND BACKGROUND

Various electrical applications such as rectifiers, limiters, clampers and many more took advantage of the diode characteristics in their operation. For example, diode rectifier is a circuit that causes an ac input voltage to be converted into a pulsed waveform having an average, or dc output voltage as Figure 2.1. To eliminate the fluctuations in the output voltage of a rectifier and produce a constant-level dc voltage; a filter usually a capacitor; is implemented in parallel with the load.

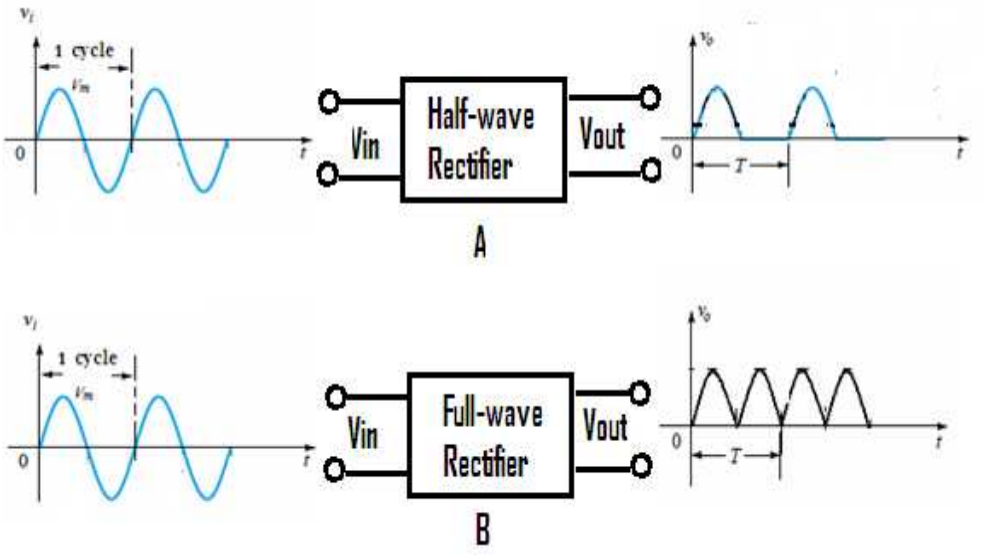


Figure 2.1: Rectifier Circuits(A) half-wave (B)Full-wave

Rectifiers Useful Formulas

Parameter	Half-wave	Center- tap	Bridge
Vp(out)	$V_{p(out)} = V_{in} - V_B$	$V_{p(out)} = V_s - V_B$	$V_{p(out)} = V_{sec} - 2V_B$
Output frequency F(out)	$F_{out} = F_{in}$	$F_{out} = 2F_{in}$	$F_{out} = 2F_{in}$
V_{dc}(without filter)	$V_{Dc} = \frac{V_{p(out)}}{\pi}$	$V_{DC} = \frac{2V_{p(out)}}{\pi}$	$V_{DC} = \frac{2V_{p(out)}}{\pi}$
V_{dc} (with filter)		$V_{DC} = \left[1 - (\frac{1}{2F_{(out)}RC})\right] * V_{p(out)}$	
Ripple factor (r)		$r = \frac{V_{r(p-p)}}{V_{DC}}$	
Power Dissipation(P_Z)		$P_Z = I_Z V_Z$	
Regulation Percentage ($\%V_O$)		$\%V_O = \left[(V_{NL} - V_{FL})/V_{FL}\right] * 100\%$	

PROCEDURE

- **Half-wave Rectifier**
  - ***Do not plug the transformer into the mains until you are told to do so.***

1. If Vsec applied to the circuit shown in Figure 2.3 is 12Vrms/50Hz, then determine and draw to scale its value in V_{p-p} on a graph paper.
2. Calculate V_{pout}, V_{DC} and F_{out}.
3. If a 220 **µF** capacitor is connected in parallel with R_L, calculate V_{DC},V_{rip} and the ripple factor.
4. If the 220 µF is replaced by 4.7 µF and 100 µF. Will the V_{DC},V_{rip} and ripple factor be affected. Explain your answer?
5. Connect the circuit and make sure the supervisor approve your connected circuit.
6. Using the Oscilloscope measure and show : V_{in} at point A, V_{pout} at point B, and V_{DC} .
7. Unplug the transformer. Connect a **220µF** capacitor in parallel with R_L . ***Pay attention to capacitor polarity***.
8. Measure V_{DC}, V_{ripp} and Calculate ripple factor.
9. Repeat steps 7, 8 for **100µF , 4.7 µF** capacitors.
10. Record your results in table 2.1.

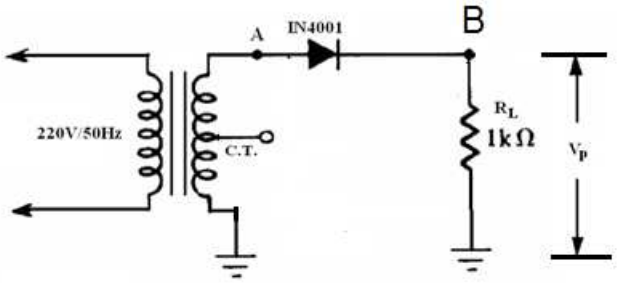


Figure 2.3: Unfiltered Half-wave Rectifier

- **Center-Tapped Full-wave Rectifier**

1. Wire the circuit shown in Figure 2.5 and repeat part one steps.

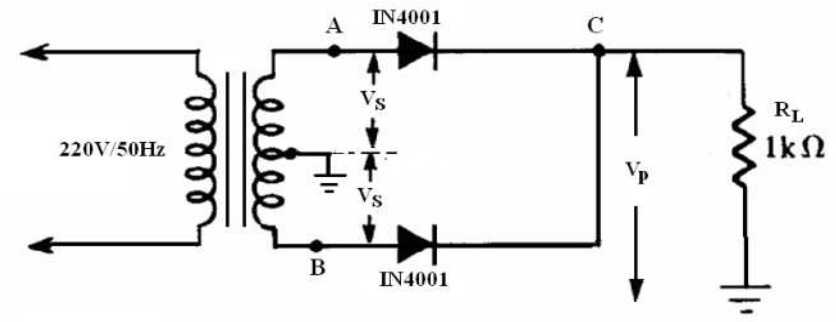


Figure 2.5: Center- Tapped Full wave Rectifier

- **Bridge Full-Wave Rectifier**

1. Wire the circuit shown in Figure 2.6 and repeat part one steps.

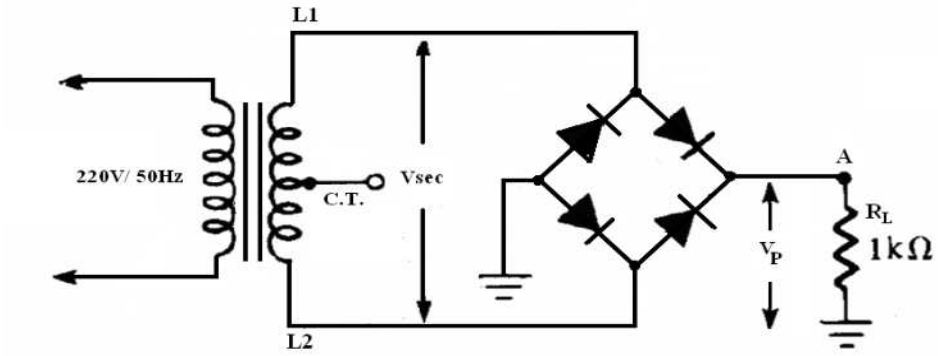


Figure 2.6: Bridge Full wave Rectifier

Table 2.1: data for Rectifier Circuits

Capacitor Value	Parameter	Rectifier Type			Comment
		H.W	C.T	Bridge	
4.7µF	V_{DC}				
	$V_{r(p-p)}$				
	Ripple Factor				
100µF	V_{DC}				
	$V_{r(p-p)}$				
	Ripple Factor				
220µF	V_{DC}				
	$V_{r(p-p)}$				
	Ripple Factor				

Summarize and compare your results.

EXPERIMENT THREE

THE DIODE LIMITER

OBJECTIVE:

- ☒ To demonstrate the operation of various diodes Limiter circuits.

EQUIPMENTS AND COMPONENTS

- ☒ **Dual Trace Oscilloscope**
- ☒ **DMM** — **2 IN 4001 Diodes**
- ☒ **Dual DC Power Supply.** — **15KΩResistors**
- ☒ **Breadboard**
- ☒ **Function Generator** — **Jumpers.**

THEORY AND BACKGROUND

Clippers (Limiters) have the ability to limit "clip-off" a portion of the input signal without distorting the remaining part of the alternating waveform. The half-wave rectifier is an example of the series configuration of diode clipper. Depending on the direction of the diode, the positive or negative region of the input signal is “clipped” off. There are two general categories of clippers: *series* and *parallel.* The series configuration is defined as one where the diode is in series with the load, while the parallel variety has the diode in a branch parallel to the load. Figure 4.1 shows positive parallel limiter.

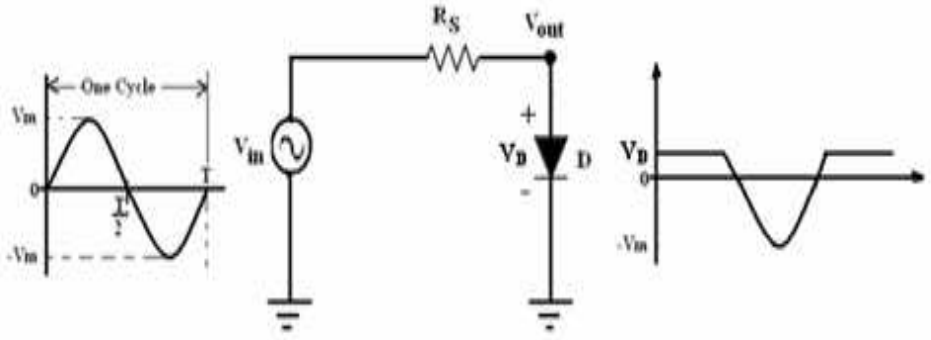


Figure 4.1: Parallel Limiter

The addition of a dc supply such as shown in Figure 4.2 can have a pronounced effect on the output of a clipper, and is known as biased limiter. V_{DC} in the voltage divider method is given by:

$$V_{DC} = \left(\frac{R_2}{R_1 + R_2} \right) * V_{CC}$$

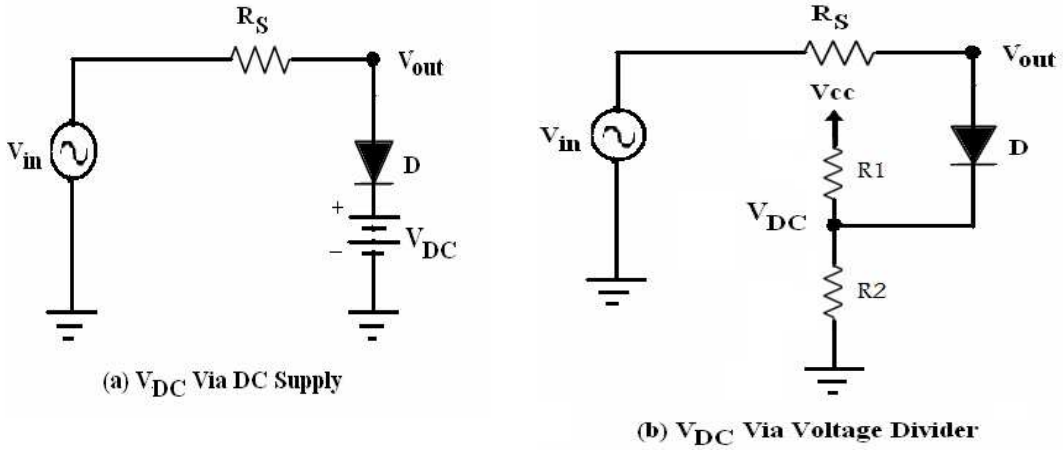


Figure 4.2: Biased Parallel limiters

PROCEDURE

1. Wire the circuit shown in Figure 4.5.
1. Connect CH1 at point A with respect to ground.
2. Adjust V_{in} to 6Vp-p/200Hz sine wave.
3. Connect CH2 at point B with respect to ground to measure the limiting level.
4. On a graph draw both input and output signals.
5. Wire the circuit of figure 4.6 . Gradually increase V_{DC} and observe V_{out} . Explain your results and determine V_{DC} value that will retain unclipped V_{out}.

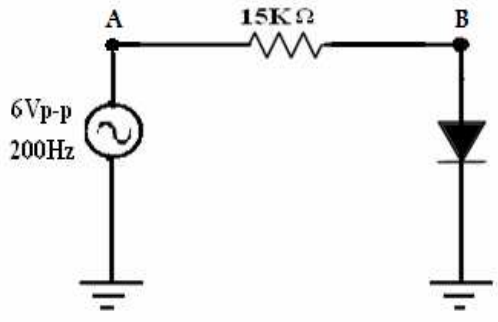


Figure 4.5: Basic Positive Limiter

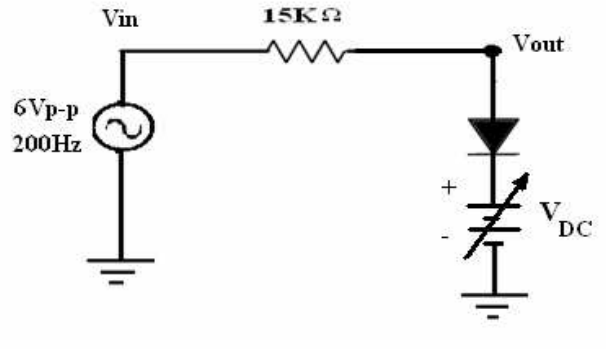


Figure 4.6: Biased Limiter

6. Wire the circuit of figure 4.7 and repeat steps 4 and 5.

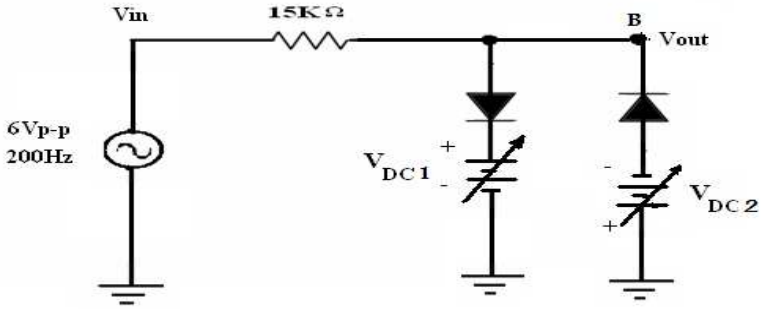


Figure 4.7: Double Biased Limiter

EXPERIMENT FOUR

THE DIODE CLAMPER

OBJECTIVE:

- ☒ To demonstrate the operation of clamper circuits.

EQUIPMENTS AND COMPONENTS

- ☒ **Dual Trace Oscilloscope**
- ☒ **DMM**
- ☒ **Dual DC Power Supply.**
- ☒ **Breadboard**
- ☒ **Function Generator**

IN 4001 Diode
10KΩ Resistor
10µF Capacitor
Jumpers.

.

THEORY AND BACKGROUND

For a clamping circuit at least three components a diode, a capacitor and a resistor are required. Sometimes an independent dc supply is also required to cause an additional shift. The important points regarding clamping circuits are:

1. The shape of the waveform will be the same, but its level is shifted either upward or downward.
2. There will be no change in the peak-to-peak or rms value of the waveform due to the clamping circuit. This is shown in the figure 4.3.
3. The values of the resistor R and capacitor C affect the waveform, and should be determined from the time constant equation of the circuit, t = RC. This value must be large enough to make sure that the voltage across the capacitor C does not change significantly during the time interval the diode is non-conducting. In a good clamper circuit, the circuit time constant t = RC should be at least ten times the time period of the input signal voltage.
4. Clamping circuits are often used in television receivers as dc restorers.

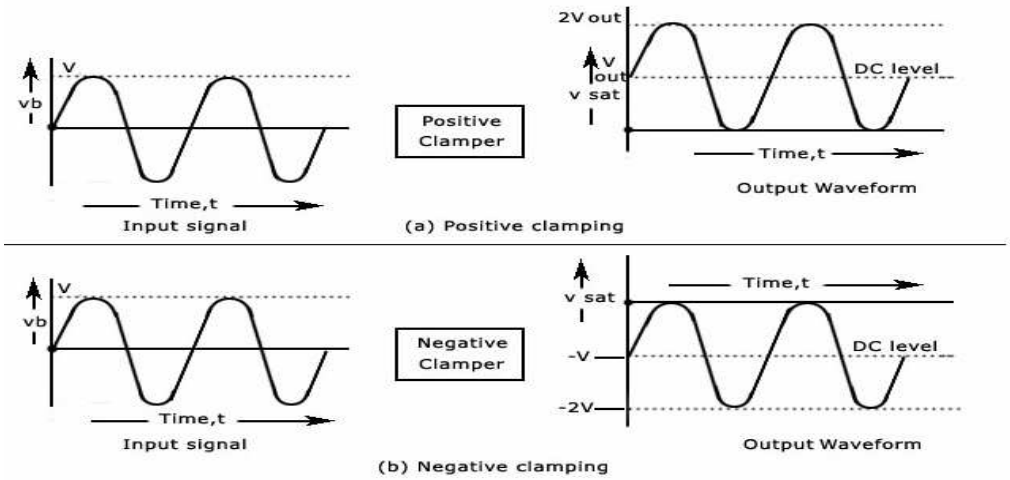


Figure 4.3: Positive clamping and Negative Clamping

Calculation for a clamper is carried out upon each half cycle of the input signal. Figure 4.4 shows a positive clamper, and the capacitor will charge during the negative half cycle, therefore:

$$V_C = V_{max} - V_D \tag{1}$$

$$V_{RL} = -V_D \tag{2}$$

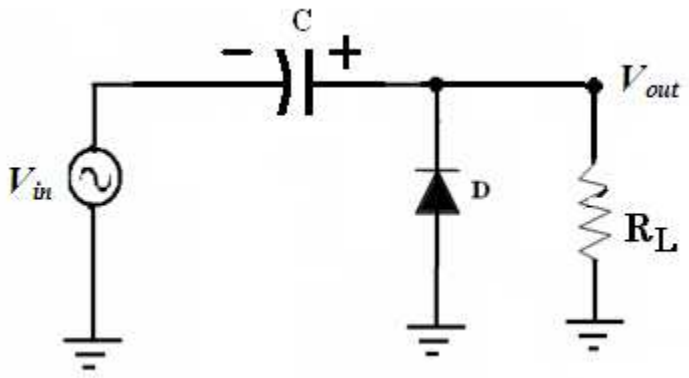


Figure 4.4: Positive Clamper

$$I_L = -\frac{V_D}{R_L} \tag{3}$$

The capacitor will discharge during the positive half cycle through R_L, and Vout is :

$$V_{out} = V_{in} + V_C \tag{4}$$

But: $V_C = V_{\max} - V_D$, then

$$V_{out} = 2V_{in} - V_D \quad (5)$$

PROCEDURE

Clamper Circuits

1. Wire the circuit shown in Figure 4.8.

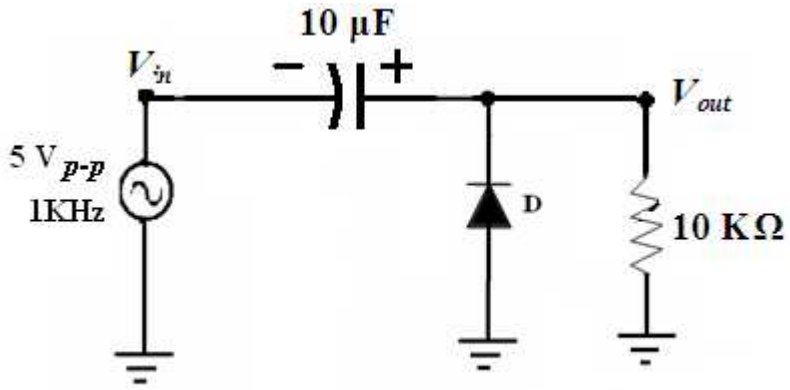
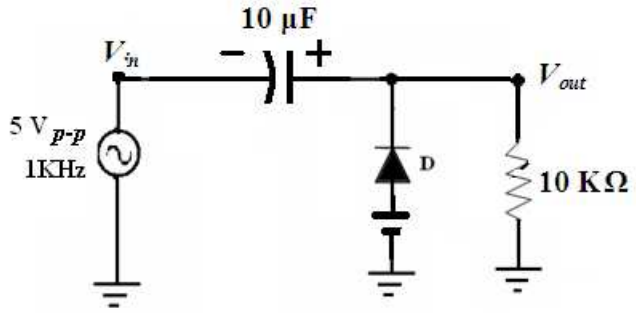


Figure 4.8: Postive Clamper

2. Turn on the function generator and adjust vin to 5Vp-p/1KHz sine wave using ch1/ac coupling of the oscilloscope.
3. Set CH 2 to Dc coupling and measure V_{out} with respect to ground . draw on the same graph both input and output signal. What will happen if CH2 is an Ac coupling.
4. If a 2V is add as shown in figure 4.9, then draw the output and explain your results.



5. Turn the power off. Reverse the diode and the capacitor.
6. Repeat steps 2 and 3.

EXPERIMENT FIVE

ZENER CHARACTERISCTICS &ZENER AS A VOLTAGE REGULATOR

Objectives

- ☒ To demonstrate the zener diode characteristics in DC mode.
- ☒ To demonstrate the operation of zener diode as a voltage regulator.

Tools and Equipments Required

- ☒ **Function Generator** — **6.2V Zener Diode.**
- ☒ **Dual Trace Oscilloscope**
- ☒ **DMM (Digital Multi Meter)** — **(2* 220Ω), 100Ω , 10Ω resistors**
- ☒ **Breadboard**
- ☒ **Dual Dc power supply**

THEORY AND BACKGROUND

Zener Diodes are basically the same as the standard PN junction diode but are specially designed to have a low pre-determined reverse breakdown voltage as shown in figure 1.1 that takes advantage of this high reverse voltage. Zener diodes are used as voltage regulators.

It had been seen that the zener diode has a region in its reverse bias characteristics of almost a constant negative voltage regardless of the value of the current flowing through the diode and remains nearly constant even with large changes in current as long as the zener diodes current remains between the breakdown current $\mathbf{I_{Z(min)}}$ and the maximum current rating $\mathbf{I_{Z(max)}}$**.** The ability of controlling the voltage can be used to great effect to regulate or stabilize a voltage source against supply or load variations, and the function of a regulator is to provide a constant output voltage to a load connected in parallel with it in spite of the ripples in the supply voltage or the variation in the load current and the zener diode will continue to regulate the voltage until the diodes current falls below the minimum $I_{Z(min)}$ value in the reverse breakdown region.

Figure 4.1 is an analysis circuit for zener diodes when are used as a voltage regulators. When E_S is less than Vz, zener acts as an open circuit and V_O is given by:

$$V_O = (R_L / (R_L + R_S) E_S \quad (1)$$

And, $$I_L = V_L / R_L \quad (2)$$

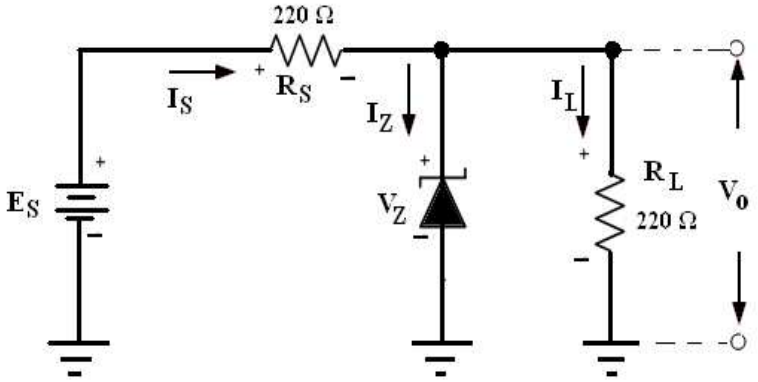


Figure 4.1: Analysis diagram for Zener diode as a regulator

When $E_S \geq V_Z$, then:

$$V_O = V_L = V_Z \quad (3)$$

And, $$I_S = I_Z + I_L \quad (4)$$

The suorce current (I_S) is given by:

$$I_S = (E_S - V_O) / R_S \quad (5)$$

And power dissipation(P_Z)is:

$$P_Z = I_Z V_Z \quad (6)$$

Regulation percentage (%V_O)

$$\%V_O = \left[(V_{NL} - V_{FL}) / V_{FL} \right] * 100\% \quad (7)$$

Where, V_{NL} — Output voltage without Load , and V_{FL} — Output voltage with Load

PROCEDURE

PART 1: ZENER CHARACTERISTICS

3. Wire the network of figure 4.2.

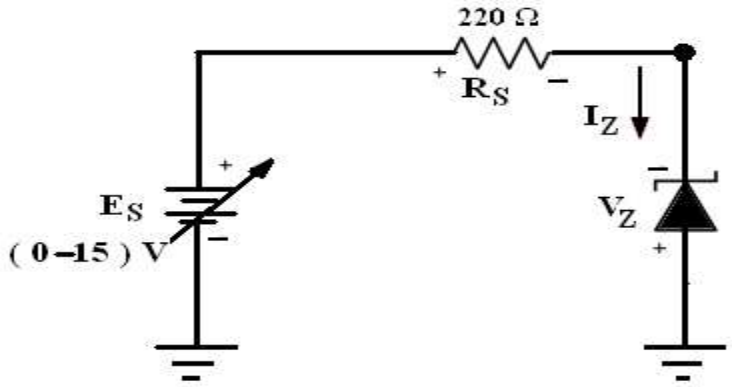


Figure 4.2: Zener Characteristics

4. Increase the dc supply voltage in small steps while simultaneously measuring the voltage across (**Vz**) and calculate the current (**Iz**) using ($I_Z = V_{RS}/R_S$) through the zener diode. In the vicinity of the zener's knee voltage (approximately 6 V), make these steps approximately 0.05V ***Do not exceed a zener current of 40 mA.*** Record your data in Table 4.2.
5. Plot your results for the corresponding zener current and voltage values on the graph provided for this purpose.
6. At I_Z =20mA, determine R_Z,

Table 4.1: Zener Characteristics

E_S (V)	V_{OUT} (V)	$I_Z = V_{RS}/R_S$ (mA)
5		
5.5		
6.00		
6.05		
6.10		
6.15		
6.20		
6.25		
6.30		
6.35		
6.40		
6.45		
6.55		
6.60		
7.00		
8.00		
9.0		
10		
11		
12		
13		
14		
15		

1- Wire the circuit shown in figure 4.3. calculate Iz, I_L and Vo. Record your results in table 4.2.

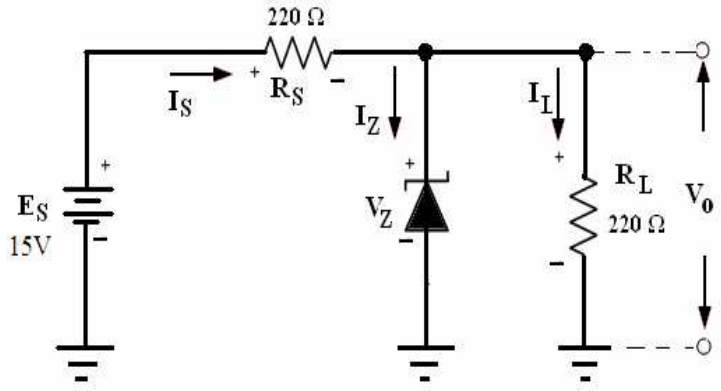


Figure 4.2: Analysis diagram for Zener diode as a regulator

2- Remove R_L, and calculate V_O.

3- Wire the circuit of figure 3.2. Measure I_S, I_Z, I_L, and V_O. record your results in table 3.1.
4- Remove R_L, and measure V_O.
5- Calculate the Regulation percentage.

Regulation percentage ($\%V_O$) = **……**

6- Explain your results.

Table 4.2: data for Zener as Voltage Regulator

Parameter	Expected value	Measured value	% error
Is			
Iz			
I_L			
V_O (with Load)			
V_O (No Load)			

7- Replace Es by an Ac input voltage as shown in figure 4.3.
8- Set v_{in} to (10vp/100Hz) . Draw the output and explain the results.

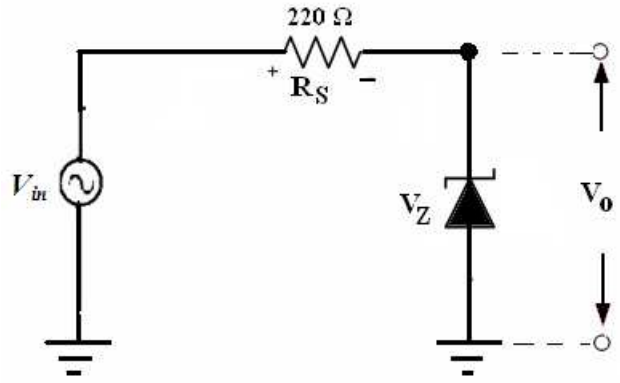


Figure 4.3: Zener With Ac input

EXPERIMENT SIX

BJT OPERATING REGIONS
&
SELF- BIASED CONFIGURATION

OBJECTIVES:

- To demonstrate the BJT operating regions the saturation , active and the cutoff regions.
- To demonstrate the fixed bias configuration as one of the transistor configuration methods.

THEORY

When operating the transistor, a dc biasing is necessary to establish the proper region of operation for ac amplification. The emitter layer is heavily doped, the base lightly doped, and the collector only lightly doped. The term bipolar reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material. If only one carrier is employed (electron or hole), it is considered a unipolar device.

Transistor Biasing is the process of setting a transistors DC operating voltage or current conditions to the correct level so that any AC input signal can be amplified correctly by the transistor. A transistors steady state of operation depends a great deal on its base current, collector voltage, and collector current and therefore, if a transistor is to operate as a linear amplifier, it must be properly biased to have a suitable operating point. Establishing the correct operating point requires the proper selection of bias resistors and load resistors to provide the appropriate input current and collector voltage conditions. The correct biasing point for a bipolar transistor, either NPN or PNP, generally lies somewhere between the two extremes of operation with respect to it being either “fully-ON” or “fully-OFF” along its load line. This central operating point is called the “Quiescent Operating Point”, or Q-point for short.

When a bipolar transistor is biased so that the Q-point is near the middle of its operating range, that is approximately halfway between cut-off and saturation, it is said to be operating as a Class-A amplifier. This mode of operation allows the output current to increase and decrease around the amplifiers Q-point without distortion as the input signal swings through a complete cycle. In other words, the output current flows for the full 360° of the input cycle.

The fixed-bias circuit of Figure 5.1 provides a relatively straightforward and simple introduction to transistor dc bias analysis. Even though the network employs an npn transistor, the equations and calculations apply equally well to a pnp transistor configuration merely by changing all current directions and voltage polarities.

EQUIPMENTS AND REQUIRED PARTS

- ☒ **Dual DC power Supply**
- ☒ **DMM**
- ☒ **Breadboard**
- ☒ **Function Generator**
- ☒ **Dual Trace Oscilloscope.**

1KΩ, 10KΩ, 100KΩ, 560KΩ
2N3904 npn silicon Transistor
Jumpers

PROCEDURE:

Part A: The BJT Operating Regions

1. Construct the circuit of Figure 5.1. Adjust $\mathbf{V_{BB}}$ to 0V. This will set I_B = Zero,
2. Set V_{CC} to 5V, calculate and then measure I_C and V_{CE} . record your results in table 5.1.
3. Set I_B to 30μA, calculate and then measure I_C and V_{CE}..
4. Set I_B to max by setting VBB to 6V, and still maintaining V_{CC} to 5V. Calculate and measure I_C and V_{CE}.

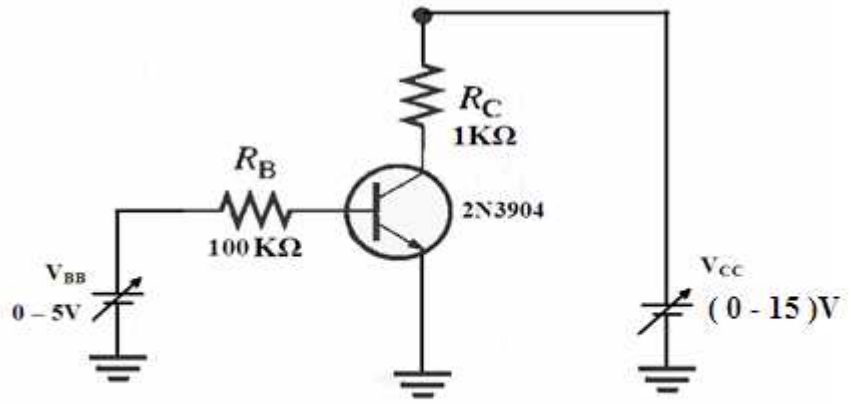


Figure 5.1: Collector Output Characteristics

Table 5.1: Collector Output Characteristics

I_B (μA)	Calculated		Measured	
	V_{CE}(V)	I_C(mA)	V_{CE}(V)	I_C(mA)
0				
30				
max				

5. Replace V_{BB} by (10Vp /100Hz) and V_{CC} to 5V as shown in figure 5.2.
6. Show and draw the output signal. Explain your results.

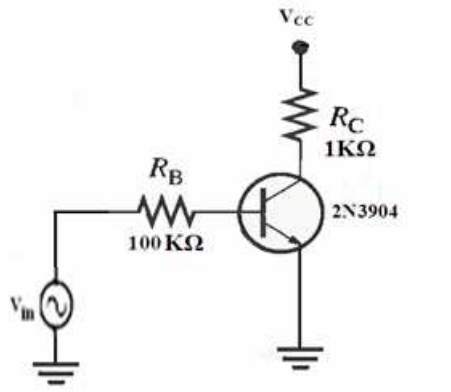


Figure 5.2: BJT Operating Regions

<u>Part B:</u> Fixed-Biased Configuration

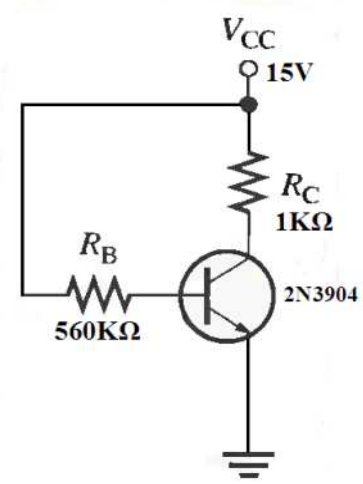


Figure 5.3: Fixed-Bias Configuration

1. Construct circuit as of Figure 5.3 using 2N3904 transistor and set $\mathbf{V_{CC}}$ = 15 V.
2. Measure the voltages $\mathbf{V_B}$ and $\mathbf{V_{RC}}$, and then record them in table 5.2.
3. Calculate the resulting base current, $\mathbf{I_B}$ and collector current, $\mathbf{I_C}$. Using the values obtained, find $\boldsymbol{\beta}$.
4. Using the values obtained in Step 3; calculate the values of $\mathbf{V_B}$, $\mathbf{V_C}$, $\mathbf{V_E}$ and $\mathbf{V_{CE}}$.
5. Heat up the transistor in Figure 5.2, and then measure $\mathbf{V_B}$, $\mathbf{V_C}$, $\mathbf{V_E}$ and $\mathbf{V_{CE}}$.
6. How do the measured values (Step 5) compare to the calculated values (Step 4)?

Table 5.2: Self- Biased Configuration Results

Parameter	Transistor		Notes
	Expected Value	Measured Value	
$\mathbf{I_B}$			
$\mathbf{I_C}$			
$\boldsymbol{\beta_{DC}}$			
$\mathbf{V_{BE}}$	**0.7V typical**		
$\mathbf{V_{CE}}$			

EXPERIMENT SEVEN

VOLTAGE DIVIDER- BIAS CONFIGURATION
&
COMMON EMITTER AMPLIFIER

OBJECTIVE

- To demonstrate voltage-divider-bias configuration.
- To investigate the operation of a common-emitter amplifier, and what influences its voltage gain.

THEORY

The things you learned about biasing a transistor in previous experiment is now applied in this experiment where bipolar junction transistor (BJT) circuit is used as small-signal amplifier. The biasing of a transistor is purely a dc operation. The purpose of biasing is to establish a Q-point about which variations in current and voltage can occur in response to an ac input signal. Voltage-divider-bias configuration is an improved level of stability that can be obtained by introducing a feedback path from collector to base. Although the *Q*-point is not totally independent of beta (even under approximate conditions), the sensitivity to changes in beta or temperature variations is normally less than encountered for the fixed-bias or emitter-biased configurations. Common-Emitter is categorized as small signal.

The term small-signal refers to the use of signals that take up a relatively small percentage of an amplifier's operational range. However; Power amplifiers are those amplifiers that have the objective of delivering power to a load. This means that components must be considered in terms of their ability to dissipate heat. Common-Emitter amplifier can also be considered as Class A power Amplifier. Any amplifier will be affected by the effect of load impedance (RL), and frequency considerations. Remember that $\mathbf{A_{vNL}}$ is the gain of the system without an applied load. Therefore; the loaded voltage gain of an amplifier is always less than the no-load level. Also the frequency of the applied signal can have a pronounced effect on the response of a single-stage or multistage network. At low frequencies, you shall find that the coupling and bypass capacitors can no longer be replaced by the short-circuit approximation because of the increase in reactance of these elements. At high frequencies, the stray capacitive elements associated with the active device and the network will limit the high-frequency response of the system. An increase in the number of stages of a cascaded system will also limit both the high- and low-frequency responses

.

REQUIRED PARTS AND EQUIPMENTS

- **Resistors (1/4 W), 10 K Ω, 150 Ω, 2.7 K Ω, Two 3.9 K Ω**
- **10 µF capacitor**
- **Two 2.2µF/ 25V capacitor**
- **2N3904 npn Transistor**
- **VOM or DMM**
- **DC Power Supply**
- **Function Generator**
- **Dual Trace Oscilloscope**
- **Breadboard**

Useful Formulas

Quiescent dc base voltage

$$(1)\ V_B = \left(\frac{R_2}{R_1 + R_2}\right) V_{CC}$$

Quiescent dc emitter voltage

$$(2)\ V_E = V_B - V_{BE}$$

Quiescent dc emitter current

$$(3)\ I_E = \frac{V_E}{R_{E1} + R_{E2}} \quad (I_C = I_E \text{ for large } \beta)$$

Quiescent dc collector voltage

$$(4)\ V_C = V_{CC} - I_C R_C$$

Quiescent dc collector- emitter voltage

$$(5)\ V_{CE} = V_{CC} - I_C (R_C + R_{E1} + R_{E2})$$

Voltage gain from base to collector

$$(1)\ Av = \frac{Vout}{V in}$$

$$(2)\ Av = \frac{R_C \parallel R_L}{R_{E1} + r'_e} \quad (\text{Normal Condition})$$

$$(3)\ Av = \frac{R_C}{R_{E1} + r'_e} \quad (\text{No Load})$$

$$(4)\ Av = \frac{R_C \parallel R_L}{R_{E1} + R_{E2} + r'_e} \quad (\text{No Bypass Capacitor})$$

Transistor ac resistance (at normal room temperature)

$$(5)\ r'_e \equiv \frac{26mV}{I_E}$$

PROCEDURE

1. Assume β for 2N3904 transistor is 200; calculate the theoretical values of **V_B, V_C, V_E**, V_{CE} **, I_B** and **I_C** for the network shown in Figure 6.1, and then record them in Table 6.1.

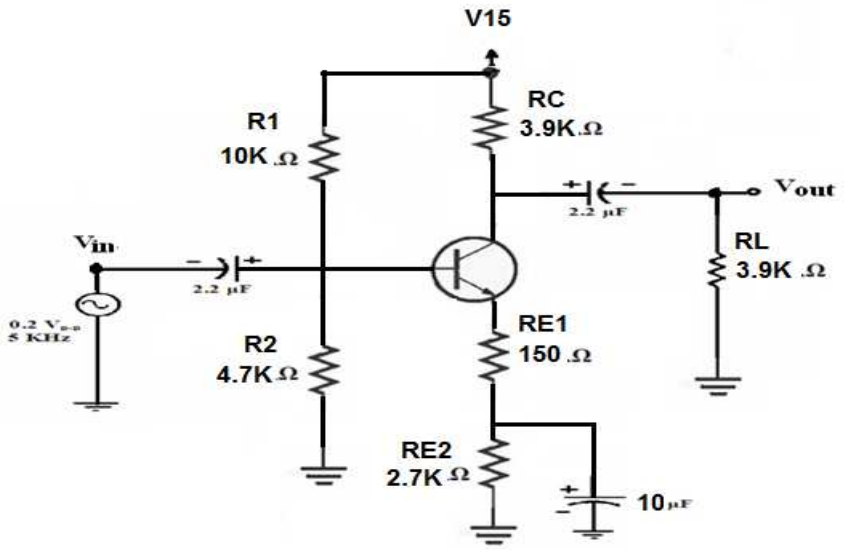


Figure 6.1: Common – Emitter Amplifier

2. Measure the parameters assigned in step one. Then calculate the % error, and Record them in table 6.1.
3. Using your measured value for Dc emitter voltage, calculate **I_E** and **re.** Record your results in table 6.1.

I_E =…….. Re=…….

Table 6.1

Parameter	Expected value	Measured value	% Error
V_B			
V_E			
V_C			
V_{CE}			
I_E			
re'			

4. Check all connections, and then set $\mathbf{V_{CC}}$ 15V and $\mathbf{V_{in}}$ to 0.2Vp-p /5KHz.
5. Connect CH1 at the input and CH2 at the output. What do you observe? Graph the both signal with proper scale on a graph paper.
6. Repeat step 5 for No Load and No Bypass Capacitor. Record your results in table 6.2.

Table 6.2

Condition	Measured		Av		% Error
	$\mathbf{V_{in}}$	$\mathbf{V_{out}}$	Expected value	Measured value	
Normal condition					
No Load					
No Bypass Capacitor					
Fin = 100 Hz					
Fin = 100 KHz					
RL 100 Ω					

EXPERIMENT EIGHT
ClASS B PUSH- PULL POWER AMPLIFIER

OBJECTIVE

- To demonstrate design and operation of Class B push- pull power amplifier.

THEORY

Class B push – pull amplifier has a pair of complementary (npn and pnp) transistors, each of which is biased at cutoff . Consequently, collector current in each transistor flows only for alternate half cycle of the input signal. Since bith transistor are biased at cutoff, the input signal must be sufficient to forward bias each transistor on the appropriate half cycle of the input signal. As a result, the crossover distortion occurs.

To eliminate crossover distortion , both transistors should be slightly forward biased so that each transistor is actually biased slightly before cutoff. This will result in a small amount of current called " trickle current".

REQUIRED PARTS AND EQUIPMENTS

- **Resistors (1/4 W), 2*10 K Ω, 3*1K Ω**
- **2* 1N4001 Silicon Diode**
- **10 µF capacitor**
- **Two 2.2µF/ 25V capacitor**
- **2N3904 npn Transistor**
- **2N3906 pnp Transistor**
- **VOM or DMM**
- **DC Power Supply**
- **Function Generator**
- **Dual Trace Oscilloscope**
- **Breadboard**

Useful Formulas

Quiescent DC collector Current (I_{CQ})

$$I_{CQ} = (V_{CC} - 2\,V_{BE}) / (R_1+R_4) \quad (\text{If } V_{BE1} = V_{BE2})$$

$$P_O\,(rms) = \{\, V_O\,(rms)\,\}^2 / R_L$$

$$P_{iDC} = V_{CC}\, I_{CQ}$$

Amplifier percent efficiency (% η)

$$\%\,\eta = P_O\,(rms) / P_{iDC}$$

PROCEDURE

1. Wire the circuit shown in figure 7.1.

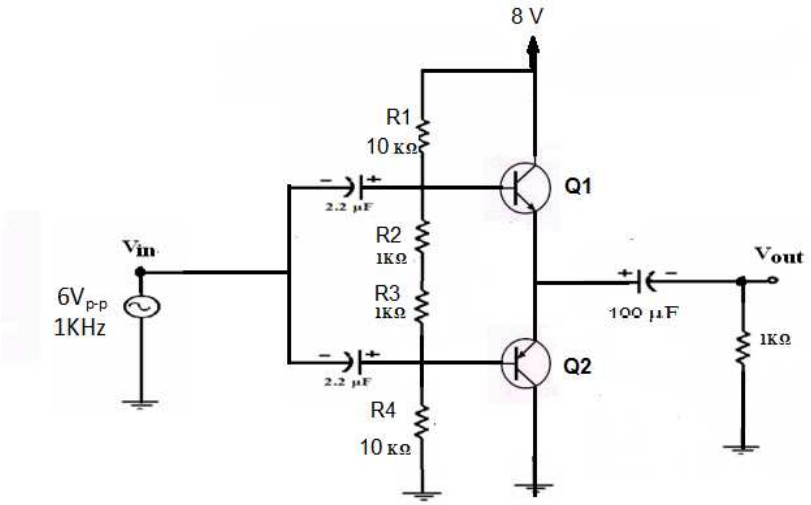


figure 7.1 : Class B Push- Pull Power Amplifier

2. Apply power to the circuit and adjust Vin to 6Vp-p/ 1KHz.
3. You should the output wave form as shown in figure 7.2.

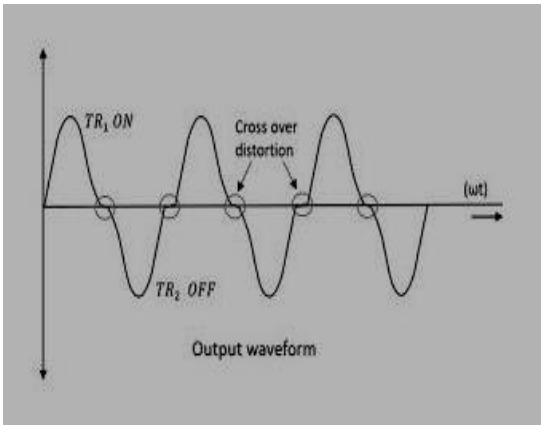


figure 7.2 : Crossover output

4. Using figure 7.2 as a guide , measure VBE for both transistors to become forward biased.
5. Turn off the power , and replace the series resistors R2, R3 with two diodes in series as shown in 7.3.
6. Observe and draw the output. Explain your results ?

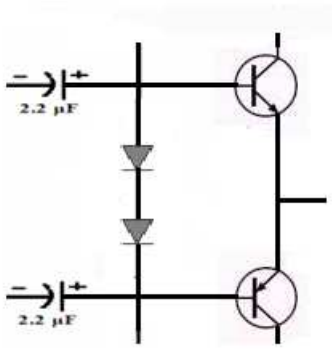


figure 7.3 : two diodes in series

7. Calculate and then measure the following shown in table 7.1

Parametre	Calculated	Measured
I_{B1}		
I_{B2}		
I_{C1}		
I_{C2}		

8. Calculate P_{iDC} , P_O (rms) and % η.

EXPERIMENT NINE
JFET CHARACTERISTICS
&
THE COMMON-DRAIN AMPLIFIER

Objective

- To determine the drain and Transfer characteristics of the JFET.
- To investigate what influences the voltage gain for the common-drain amplifier

Theory

The FET 'field-effect transistors' are unipolar devices because they operate only with one type of charge carrier. The two main types of FETs are the junction field-effect transistor (JFET) and the metal oxide semiconductor field-effect transistor (MOSFET). The FETs voltage-controlled devices, where the voltage between two of the terminals (gate and source) controls the current through the device operate with a reverse-biased pn junction to control current in a channel. JFETs fall into either of two categories, n-channel or p-channel. Figure 8.1 shows the basic structure of an n-channel JFET (junction field-effect transistor).

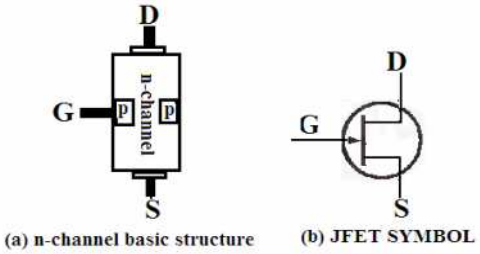


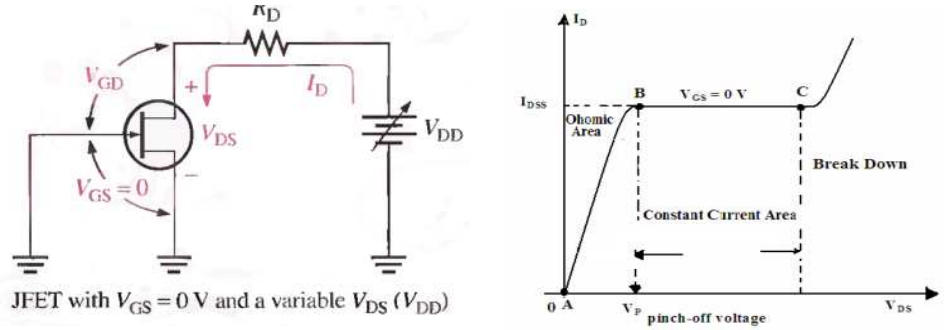
Figure 8.1: Basic Structure and Symbol

Basic Operation

If dc bias voltages are applied to an n- channel device. V_{DD} provides a drain-to-source voltage and supplies current from drain to source. V_{GG} sets the reverse-bias voltage between the gate and the source. Reverse-biasing of the gate-source junction with a negative gate voltage produces a depletion region along the pn junction, which extends into the n- channel and thus increases its resistance by restricting the channel width. The channel width and thus the channel resistance can be controlled by varying the gate voltage, thereby controlling the amount of drain current, $\mathbf{I_D}$.

JFET CHARACTERISTICS AND PARAMETERS

When the gate-to-source voltage is zero (i.e. the gate to the source is shorted, **(V_{GS} = 0 V).** As **V_{DD}** (and thus **V_{DS}**) is increased from 0 V, **I_D** will increase proportionally, as shown in the graph of Figure 8.2(b) between points A and B. In this area, the channel resistance is essentially constant because the depletion region is not large enough to have significant effect. This is called the ohmic area because V_{DS} and I_D are related by Ohm's law.



(a) (b)

Figure 8.2: JFET characteristics circuit and curve

At point B in Figure 8.2(b), the curve levels off and **I_D** becomes essentially constant. As **V_{DS}** increases from point B to point C, the reverse-bias voltage from gate to drain (**V_{GD}**) produces a depletion region large enough to offset the increase in V_{DS}, thus keeping I_D relatively constant.

Pinch-Off Voltage, V_P: is **the** value of **V_{DS}** at which **I_D** becomes essentially constant (point B) on the curve in Figure 8.2(b). For a given JFET, V_P has a fixed value

I_{DSS} (Drain to Source current with gate Shorted): is the maximum drain current that a specific JFET can produce regardless of the external circuit, and it is always specified for the condition, V_{GS} = 0 V.

Breakdown occurs at point C when I_D begins to increase very rapidly with any further increase in V_{DS}. Therefore; there is a difference between pinch-off voltage and cutoff. V_P is the value of V_{DS} at which the drain current becomes constant and is always measured at V_{GS} = 0V. Also V_{GS}(off) and V_P are always equal in magnitude but opposite in sign.

Cutoff Voltage, V_{GS}(off): is The value of V_{GS} that makes I_D approximately zero. Therefore, The JFET must be operated between V_{GS} = 0 V and V_{GS} (off).

JFET Transfer Characteristic

Figure 8.2 shows that a range of **V_{GS}** values from zero to **V_{GS}**(off) controls the amount of drain current. For an n-channel JFET,**V_{GS}**(off) is negative, and for a p-channel JFET.V_{GS}(off) is positive. Because V_{GS} does control I_D the relationship between these two quantities is very important.

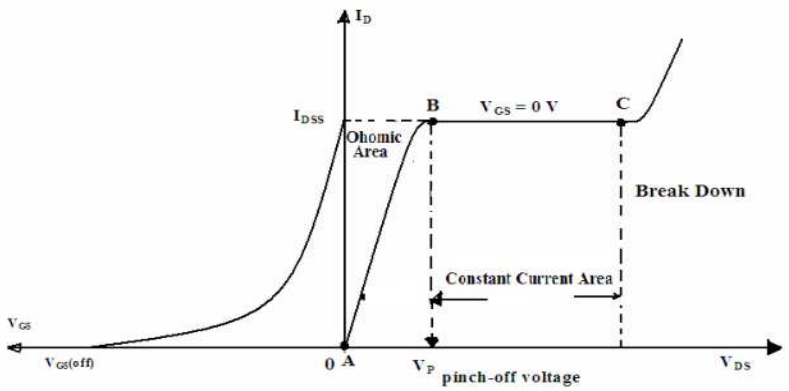


Figure 8.2: Developing of Transfer Curve

Notice that the bottom end of the curve is at a point on the V_{GS} axis equal to $V_{GS(off)}$, and the top end of the curve is at a point on the I_D axis equal to I_{DSS}. This curve, of course, shows that the operating limits of a JFET are:

$$I_D = 0 \qquad \text{when } V_{GS} = V_{GS(off)}$$

And

$$I_D = I_{DSS} \qquad \text{when } V_{GS} = 0$$

The transfer characteristic curve can also be developed from the drain characteristic curves by plotting values of I_D for the values of V_{GS} taken from the family of drain curves at pinch-off, as illustrated in Figure 8.2 for a specific set of curves. Each point on the transfer characteristic curve corresponds to specific values of V_{GS} and I_D on the drain curves. The JFET transfer characteristic curve is expressed as:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

JFET Forward Tran-conductance (g_m) : is the change in drain current (ΔI_D) for a given change in gate –to- source voltage (ΔV_{GS}) with the drain- to- source voltage constant. It is expressed as

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

Data sheets normally give the value of **(g_m)** measured at V_{GS} =0V(**g_{m0}**)

Given the value of **g_{m0} (Y_{fs}),** g_m can be calculated

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)$$

If the value of (g_{m0}) **is not available we can use the values of**

$$g_{m0} = \frac{2I_{DSS}}{|V_{GS(off)}|}$$

The common-drain amplifier, often referred to as a source-follower, is characterized by application of the input signal to the gate lead while the output is taken from the source. The output signal is never larger than the input but is always in-phase with the input. Consequently, the output follows the input.

Equipments and Components

- ☒ **Dual DC Power Supply**
- ☒ **DMM**
- ☒ **2N3819 n-channel JFET or MPF 102**
- ☒ **Function Generator**
- ☒ **Two 2.2µ F Capacitor**

Breadboard
Resistor 100 Ω,
100 kΩ Resistor
Two 1kΩ Resistor

USEFUL FORMULAS

Voltage gain

$$(1)\ A_v = \frac{v_{out}}{v_{in}}$$

$$(2)\ A_v = \frac{g_m R_s}{g_m R_s + 1}$$

where $R_s = R_S \| R_L$

JFET dc gate-to-source cutoff voltage

$$(3)\ V_{GS(off)} = \frac{2I_{DSS}}{g_{m0}}$$

Quiescent dc drain (source) current

$$(4)\ I_D = 2I_{DSS}\left[\frac{(R_S g_{m0} + 1) - (2R_S g_{m0} + 1)^{1/2}}{(R_S g_{m0})^2}\right]$$

Quiescent dc gate-to source voltage

$$(5)\ V_S = I_D R_S$$

$$(6)\ = -V_{GS} \qquad \text{so that } V_G \simeq 0$$

Quiescent dc drain-to-source voltage

$$(7)\ V_{DS} = V_{DD} - I_D R_S$$

JFET forward transconductance at Q point

$$(8)\ g_m = g_{m0}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]$$

PROCEDURE

Part One: Characteristics Curves

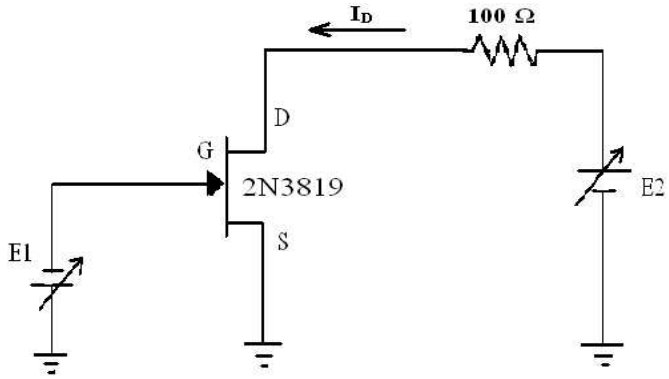


Figure 8.3: JFET Characteristics Circuit

A: Drain Characteristics

1. Construct the circuit of Figure 8.3.
2. Set $V_{GS} = 0$. ***{This value must be constant during this part.}***
3. Vary E2 (V_{DD}) to vary V_{DS} as shown in table 8.1.
4. Measure the corresponding I_D. Record your results in table 6.1.
5. Repeat steps 3,4 for **V_{GS} = -- 1V, (--2V).**
6. Graph to scale on graph paper the drain characteristics.

Table 8.1: Drain Characteristics

V_{DS} (V)	0	0.1	0.2	0.3	0.5	1	2	3	4	6	8	10	12
I_D (mA)													

B: Transfer Characteristics

1. Set $V_{DS} = 12V$. ***{This value must be constant during this part.}***
2. Vary E1 to vary V_{GS} as shown in table 8.2.
3. Measure the corresponding I_D. Record your results in table 8.2.
4. Graph to scale on graph paper the transfer characteristics. Show Vp and I_{DSS}.

Table 8.2: Transfer Characteristics

V_{GS} (V)	0	- 0.1	- 0.2	- 0.5	- 1	- 2	- 3	- 4	Vp
I_D (mA)									

Part Two: Common Drain Amplifier

1. **Using the <u>JFET parameters measured in part one,</u>** calculate the JFET's quiescent current $\boldsymbol{I_{DSS}}$ and gate-to-source voltage $\boldsymbol{V_{GS}}$ recording your values in Table 8.3. Based on the values for g_{m0} and I_{DSS} that you measured for the same JFET in this experiment, calculate the JFET's Tran-conductance g_m at this quiescent point, and record your result in Table 8.3.

2. Wire the circuit shown in Figure 8.4, omitting the signal generator and the power supply.
3. After you have checked all connections, apply only the 15-V supply voltage to the breadboard. With your VOM and DMM, measure the JFET's quiescent drain current $\boldsymbol{I_{DSS}}$ and gate-to-source voltage V_{GS} recording your values in Table 8.3. Based on the values for $\boldsymbol{g_{m0}}$ and I_{DSS} that you measured for the same JFET in this experiment, calculate the JFET's trans-conductance g_m at this quiescent point, and record your result in Table 8.3. In all cases, compare your measured values with what you would expect to measure.
4. Connect Channel 1 of your oscilloscope to the amplifier's input (v_{in}) and CH2 to the 1-kΩ load resistor **($\boldsymbol{v_{out}}$).** Then connect the signal generator to the circuit as shown in Figure 8.4, and adjust the sine wave output level of the generator at 1V peak-to-.peak at a frequency of 5 kHz.

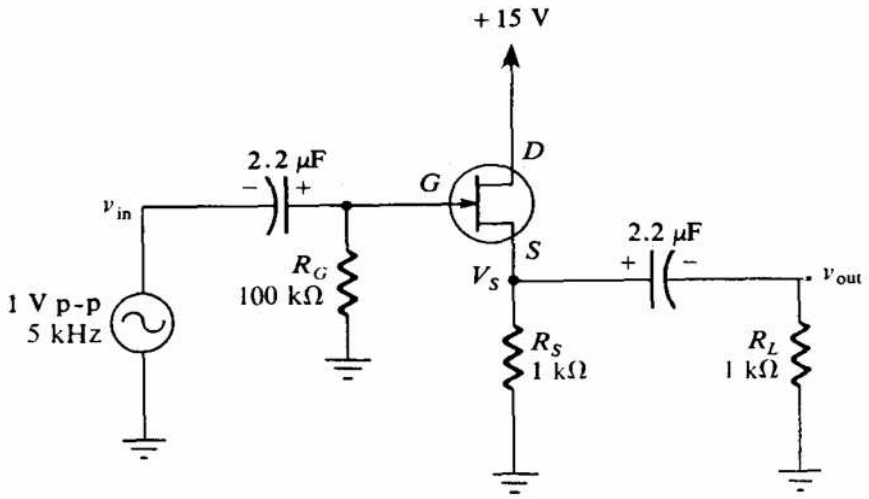


Fig. 8.4: Common Drain Amplifier

5. Note that the output signal level (v_{out}) is less than the input level ($\boldsymbol{v_{in}}$). In addition, ($\boldsymbol{v_{out}}$) is in-phase with the input. These points are two major characteristics of a common-drain amplifier.

6. Calculate the voltage gain (Equation 2) using the Tran-conductance value determined in Step 1, and record the value in Table 8.4. Now measure the actual

circuit voltage gain by dividing the peak-to-peak output voltage (v_{out}) by the peak-to-peak input voltage (v_{in}) (Equation 1), recording your result in Table 8.4.

7. Now remove RL. What do you observe? Experimentally determine the voltage gain by measuring (v_{out}) and (v_{in}), comparing your measured result with the expected value. In this case, Rs = R_S. Record your results in Table 8.4.Explain your results?

Table 8.3: FET Parameters

From Part 1 g_{m0} = I_{DSS} =			
Parameter	Measured Value	Expected Value	% Error
I_D			
$-V_{GS}$			
g_m			

Table 8.4: Common Drain Amplifier Data

Condition	v_{in}	v_{out}	Measured Gain	Expected Gain	% Error
Normal circuit					
No load					

EXPERIMENT TEN
OP- AMPS (OPERATIONAL AMPLIFIES)

OBJECTIVE

- To introduce the 741 operational amplifier.
- To demonstrate the operation of both inverting and non-inverting amplifier circuits using a74l operational amplifier.

THEORY

The operational amplifier, or op-amp, is a very high gain differential amplifier with high input impedance and low output impedance. Typical uses of the operational amplifier are to provide voltage amplitude changes (amplitude and polarity), oscillators, filter circuits, and many types of instrumentation circuits. An op-amp contains a number of differential amplifier stages to achieve a very high voltage gain. Figure 9.1 shows a basic circuit connection. The output voltage is limited by the supply voltage due the Virtual Ground. That is; if Vout = 10 V, and Av = 20000, the input voltage would be :

Vi = - Vout / Av then Vi = - 10 / 20000 = 0.5mV

.

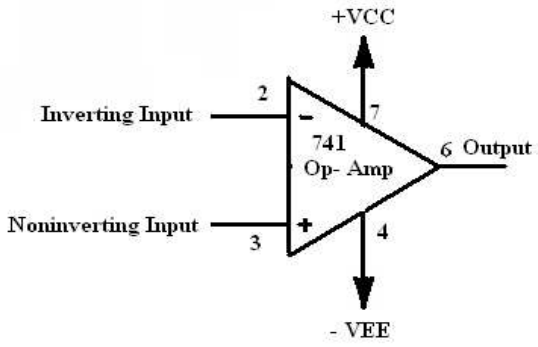


Figure 9.1: Op- Amp.Basic Connections

The inverting amplifier's closed-loop voltage gain can be equal to, or greater than 1. As its name implies, its output signal is always inverted with respect to its input signal. On the other hand, the non-inverting amplifier's closed loop voltage gain is always greater than 1, while the input and output signals are always in-phase.

USEFUL FORMULAS

Inverting amplifier closed-loop voltage gain

$$(1)\ A_v = -\frac{R_f}{R_i}$$

Noninverting amplifier closed-loop voltage gain

$$(2)\ A_v = 1 + \frac{R_f}{R_i}$$

REQUIRED PARTS AND EQUIPMENTS

- ☒ **Dual DC Power Supply**
- ☒ **Dual Trace Oscilloscope**
- ☒ **VOM or DMM**
- ☒ **Function Generator**
- ☒ **Breadboard**

741 Op-Amp.
100K Ω Resistor
2* 10 K Ω Resistor

PROCEDURE

Part One: Inverting and Non-Inerting Amplifiers

1. Calculate the expected voltage gain AV for circuit shown in figure 9.2. record your results in table 9.1.

   Av (10kΩ) = ……..
   Av (1kΩ) = ……

2. Wire the circuit shown in figure 9.2, and set your oscilloscope for the following approximate settings:

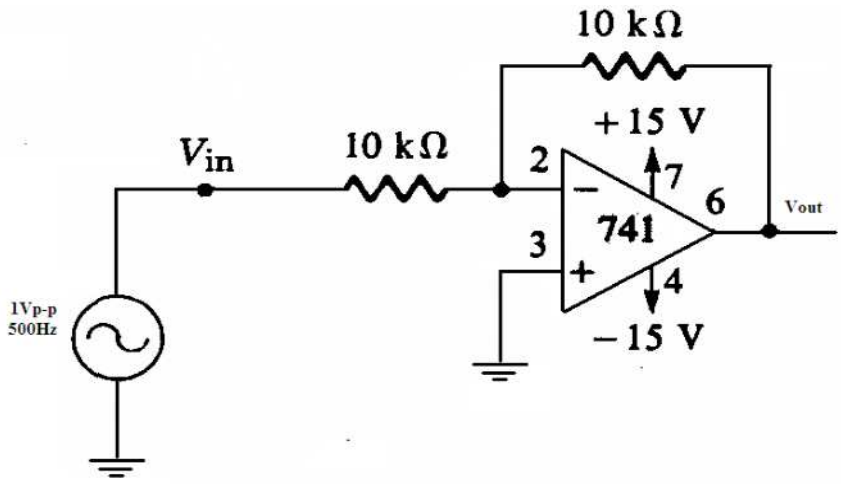


Figure 9.2: Inverting Amplifier

3. Adjust the F.G to 0.5Vp-p/500Hz.Measure the peak-to-peak output voltage, and record your results in table 9.1. Determine the voltage gain and compare it with the expected value.
4. Keeping the input signal at 1 V peak-to-peak, change resistor Rf according to Table 9.1., recording your results as in Step 3. ***Each time, disconnect the power supply and signal generator before you change the resistor.***
5. Now wire the noninverting amplifier circuit shown in Figure 9.3. Apply power to the breadboard and adjust the input voltage to 1 V peak-to-peak and the frequency at 400 H Again position the input voltage above the output voltage on the oscilloscope's display. What is the difference between the two signals?

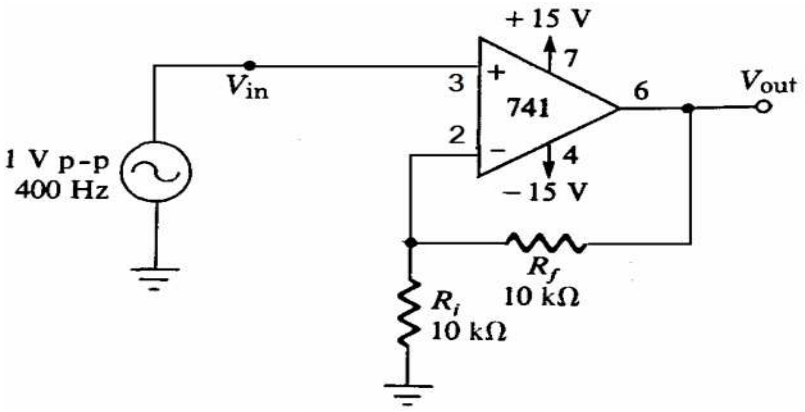


Figure 9.3: Non-Inverting Amplifier

6. Repeat steps 3 & 4. Record your results in Table 9.2.

Table 9.1: Inverting Amplifier

R_f	Measured V_{out}	Measured Gain	Expected Gain	% Error
10 kΩ				
4.7 kΩ				
100 kΩ				
1 kΩ				

Table 9.2: Non Inverting Amplifier

R_f	Measured V_{out}	Measured Gain	Expected Gain	% Error
10 kΩ				
4.7 kΩ				
100 kΩ				
1 kΩ				

Part Two: OP-AMP COMPARATORS

The purpose of this part is to demonstrate the operation of noninverting and inverting comparator circuits using a 741 operational amplifier. A comparator determines whether an input voltage is greater than a predetermined reference level. Since a comparator operates in an open-loop mode, the output voltage approaches either its positive or its negative supply voltage.

USEFUL FORMULAS

Noninverting comparator output

(1) $V_{out} = +V_{SAT}$ when $V_{in} > V_{REF}$

(2) $V_{out} = -V_{SAT}$ when $V_{in} < V_{REF}$

Inverting comparator output

(3) $V_{out} = +V_{SAT}$ when $V_{in} < V_{REF}$

(4) $V_{out} = -V_{SAT}$ when $V_{in} > V_{REF}$

PROCEDURE

1. Wire the circuit shown in the schematic diagram of Figure 7.3, and set your oscilloscope for the following approximate settings:

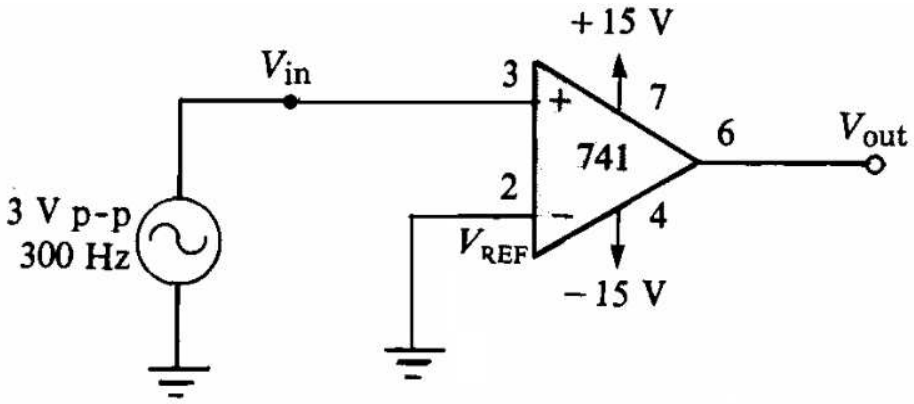


Figure 7.3: Comparator

2. Apply power to the breadboard, and adjust the input voltage at 3Vp-p, and the frequency at 300 Hz. What is the polarity of the output voltage when the input signal goes positive? When the input goes negative?
3. Disconnect the power and signal leads to the breadboard, and reverse the input connections to the op-amp so that the input signal is now connected to the inverting input while the noninverting input is grounded.

4. Again apply both the power and signal leads to the breadboard. Now what is the difference between the operation of this circuit and that of the circuit used earlier?

EXPERIMENT ELEVEN

OP-AMP DIFFERENTIATOR AND INTEGRATOR

PURPOSE AND BACKGROUND

The purpose of this experiment is to demonstrate the operation of both a differentiator and an integrator using an op-amp. A differentiator is a circuit that calculates the instantaneous slope of the line at every point on a waveform. On the other hand, an integrator computes the area underneath the curve of a given waveform. Differentiation and integration are paired mathematical operations in that one has the opposite effect of the other. For example, if you integrate a waveform and then differentiate it, you obtain the original waveform.

REQUIRED PARTS AND EQUIPMENT

Resistors (1/4 W):
- ☐ 2.2 kΩ
- ☐ Two 10 kΩ
- ☐ 22 kΩ
- ☐ 100 kΩ

Capacitors:
- ☐ 0.0022 μF
- ☐ 0.0047 μF

- ☐ 741 op-amp (8-pin mini-DIP)
- ☐ Two 0–15 V dc power supplies
- ☐ Function generator
- ☐ Dual trace oscilloscope
- ☐ Breadboarding socket

USEFUL FORMULAS

Differentiator

Output voltage:

$$(1)\quad V_{out} = -R_F C\left(\frac{dV_{in}}{dt}\right)$$

Low-frequency response:

$$(2)\quad f_c = \frac{1}{2\pi R_S C}$$

When $f_{in} < f_c$, the circuit acts as a differentiator.
When $f_{in} > f_c$, the circuit approaches an inverting amplifier with a voltage gain of $-R_F/R_S$.

Integrator

Output voltage:

$$(3)\quad V_{out} = -\frac{1}{R_1 C}\int_0^t V_{in}\,dt$$

Low-frequency response:

$$(4)\quad f_c = \frac{1}{2\pi R_S C}$$

When $f_{in} > f_c$, the circuit acts as an integrator.
When $f_{in} < f_c$, the circuit approaches an inverting amplifier with a voltage gain of $-R_S/R_1$.

For minimum output offset due to input bias currents:

$$(5)\quad R_2 = \frac{R_1 R_S}{R_1 + R_S}$$

PROCEDURE

1. Wire the differentiator circuit shown in Figure 10.1A, and set your oscilloscope to the following approximate settings:

   Channel 1: 0.5 V/division, dc coupling
   Channel 2: 0.05 V/division, dc coupling
   Time base: 0.5 ms /division

2. Apply power to the breadboard, and adjust the peak-to-peak voltage of the input triangle wave at 1 V and the frequency at 400 Hz. What is shape of the output signal?

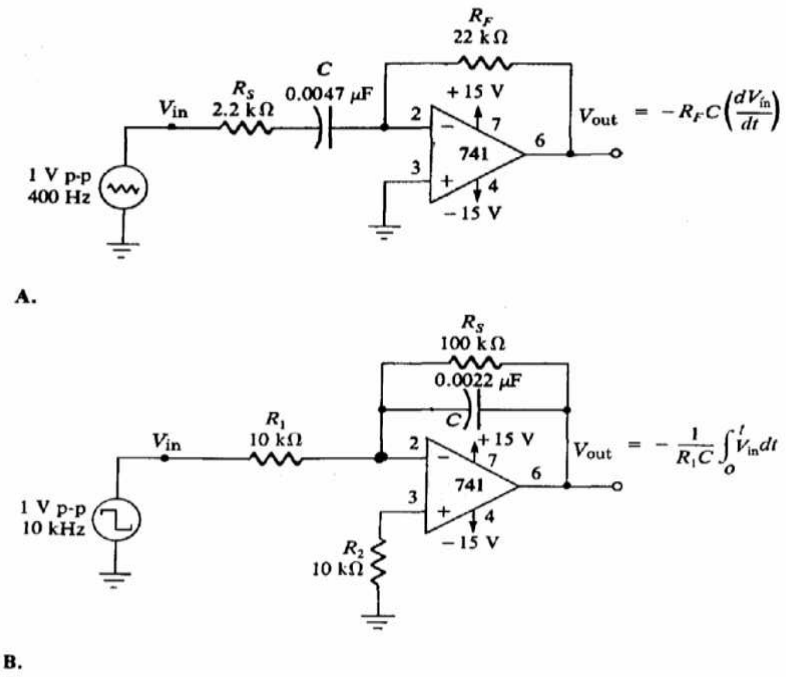


Figure 10.1: (a) Differentiator Circuit (b) Integrator

3. Temporarily remove the probe connected to Channel 2 of the oscilloscope, and adjust the resulting straight line (ground level) at some convenient position on the screen. Reconnect the probe to the output of the differentiator and measure the negative peak voltage (with respect to ground) of the square wave, recording your result in Table 10.1

4. Change the time base to 0.2 ms/division and Channel 2 to 0.1 V/division. Then adjust the input frequency at1kHz. Repeat Steps 2 and 3. You should find that the peak output voltage increases

5. Now change the input frequency to 30 kHz. Adjust the time base to 10 µs/division and Channel 2 to 2V/division. What does the output signal look like?

6. Measure the peak-to-peak output voltage and determine the voltage gain, recording your values in Table 10.1. How does the voltage gain compare to that of an inverting amplifier?

7. Wire the integrator circuit shown in the schematic diagram of Figure 10.1.B, and set your oscilloscope to the following approximate settings:

   Channels 1 and 2: 0.5 V/division, dc coupling
   Time base: 20 us/division

8. Apply power to the breadboard, and adjust the peak-to-peak voltage of the input square wave at 1 V and the frequency at 10 kHz. How does the output signal look like? Is there a phase shift between the input and output signals? Draw the waveforms? Record your results in Table 10.2.
   …….

   …….

9. Change the time base to 50 µs/division and Channel 2 to 1 V/division. Then adjust the input frequency to 4 kHz. Repeat Step8.You should find that the peak output voltage increases. does it?

   ……..

   ……

10. Now change the input frequency to 100 Hz. Adjust the time base to 2 ms/division and Channel 2 to 5 V/division. What does the output signal look like? Notice that the output signal looks like a square wave with a phase shift of 180°. Why?

    ……
    ……

11. Measure the peak-to-peak output voltage and determine the voltage gain, recording your values in Table 10.2. How does the voltage gain compare to that of an inverting amplifier?

……
……

Table 10.1: differentiator

Input Frequency	Measured Peak Output	Expected Peak Output	% Error
400Hz			
1KHz			
30KHz			

Table 10.2: Integrator

Input Frequency	Measured Peak Output	Expected Peak Output	% Error
10KHz			
4KHz			
100Hz			

EXPERIMENT TWELVE

THE PHASE-SHIFT OSCILLATOR

PURPOSE AND BACKGROUND

The purpose of this experiment is to demonstrate the design and operation of an op-amp phase-shift oscillator. By providing three RC networks having a total phase shift of 180 as positive feedback to the input of a total phase shift of the amplifier and that of the phase-shift network is 0, and the loop gain is unity. The sinusoidal output of the oscillator has a peak-to-peak voltage equal to the difference between the Op amp positive and negative saturation voltages.

REQUIRED PARTS AND EQUIPMENT

Resistors (1/4 W):
- ☐ Three 1 kΩ
- ☐ 27 kΩ

☐ 5-kΩ potentiometer
☐ Three 0.1-μF capacitors
☐ 741 op-amp (8-pin mini-DIP)
☐ Two 0–15 V dc power supplies
☐ Dual trace oscilloscope
☐ Breadboarding socket

USEFUL FORMULAS

Output frequency

$$(1)\ f_o = \frac{1}{2\pi RC\sqrt{6}}$$

For oscillation

$$(2)\ R_f/R = 29$$

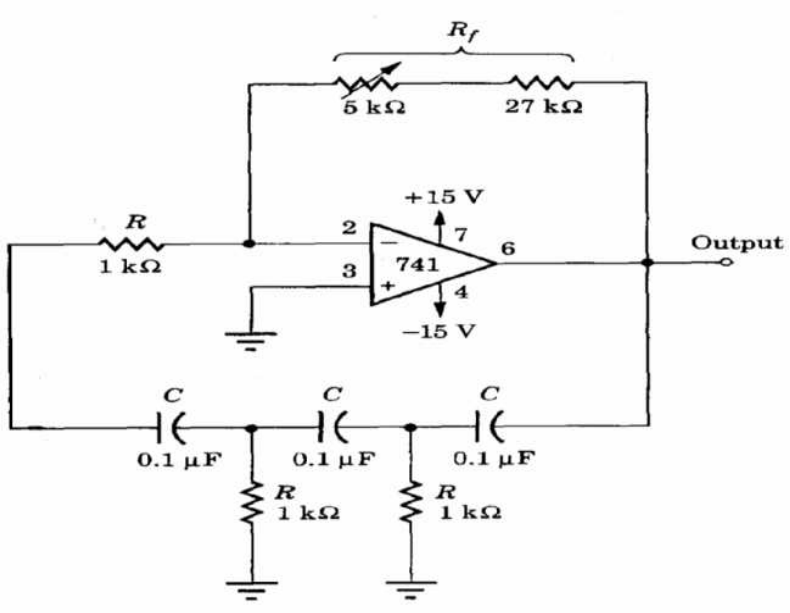


Figure 11.1: Phase Shift Oscillator

PROCEDURE

1. Wire the circuit shown in Figure 11.1, and set your oscilloscope to the following approximate settings:

   Channel 1: 5 V/division, ac coupling
   Time base: 0.5 ms /DIV
2. After you have checked all connections, apply the ± 15-V power supply connections to the breadboard.
3. Depending on the setting of the 5-kΩ potentiometer, the circuit may or may not be oscillating when power is applied. If a sine wave is not displayed on the oscilloscope, carefully adjust the 5-kΩ potentiometer until a sine wave starts to appear on the oscilloscope display. If you continue to increase the resistance of the potentiometer, you should observe that the peaks of the sine wave become clipped and that it becomes lower. Adjust the potentiometer to the point at which the circuit just sustains oscillation. On the other hand, if a sine wave is seen when power is applied on the breadboard, carefully decrease the resistance of the potentiometer to obtain the best-looking sine wave.
4. Using your oscilloscope's time base set at approximately 0.2 ms/division; measure the output frequency of the phase-shift oscillator, recording your result in Table 9.1. Compare this value with the expected frequency found using Equation 1 given in the Useful Formulas section of this experiment.
5. Disconnect the power from the breadboard and carefully remove and measure the total resistance (**R*f***) of the 27-kΩ resistor and of the setting of the series 5- kΩ potentiometer that produced oscillation. Record the value in Table 11 - 1. At the oscillation frequency set by the three RC networks, 1/29 of the output signal is fed

back to the input of the op-amp. For the loop gain to be unity, the voltage gain of the inverting amplifier must then be 29, which implies that the feedback resistor (**R*f***)must be equal to 29R How does the sum of the 27- kΩ resistor an setting of 5-kΩ potentiometer compare with the 1-kΩ resistors of the phase-shift network?

Table 11.1

Parameter	Measured	Expected	% Error
Output Frequency , *fo*			
R_f		29KΩ	

EXPERIMENT THERTEEN

THE BUTTERWORTH 2ND-ORDER LOW-PASS ACTIVE FILTER

PURPOSE AND BACKGROUND

The purpose of this experiment is to demonstrate the operation and characteristics of a Butterworth Sallen and Key 2nd-order low-pass active filter. A Butterworth low-pass filter passes all signals with frequencies below its cutoff frequency with a constant, or maximally flat, passband gain. The cutoff frequency is also referred to as the critical corner, break, or 3-dB frequency. Above this frequency, the input signal is attenuated at a rate of -12 dB/octave, a rate that is equivalent to – 40 dB/decade for such a 2nd-order filter. Because of the component values used in this experiment, the passband voltage gain is ideally fixed at 1.586 (4 dB), although other arrangements allowing other higher passband gains are possible. In addition, for input frequencies well below the cutoff frequency, there is no phase shift between input and output signals. active filter. For a Butterworth high-pass filter has an operation that is opposite that of a low-pass filter. That is, a high-pass filter passes all signals with frequencies above its cutoff frequency with a constant, or maximally flat, passband gain. Below this frequency, the input signal is attenuated at a rate of 12 dB/octave, or 40 dB/decade for such a 2nd-order filter. Because of the component values used in this experiment, the passband voltage gain is fixed at 1.586 (4 dB), although arrangements allowing other higher passband gains arc possible. In addition, for frequencies well above the cutoff frequency, there is no phase shift between input and output signals.

REQUIRED PARTS AND EQUIPMENT

Resistors (1/4 W):
- ☐ Two 6.8 kΩ
- ☐ 27 kΩ
- ☐ 47 kΩ

☐ Two 0.033-μF capacitors

☐ 741 op-amp (8-pin mini-DIP)
☐ Two 0–15 V dc power supplies
☐ Signal generator
☐ Dual trace oscilloscope
☐ Breadboarding socket

USEFUL FORMULAS

(1) $R_1 = R_2$ and $C_1 = C_2$

(2) $R_B = 0.586\, R_A$

Cutoff frequency

(3) $$f_c = \frac{1}{2\pi R_1 C_1}$$

dB frequency response

(4) $$A_{dB} = 20 \log \left(\frac{V_{out}}{V_{in}} \right)$$

(5) $$A_{dB} = 20 \log \left(\frac{1.586}{\left[1 + \left(\frac{f_{in}}{f_c} \right)^4 \right]^{1/2}} \right)$$

PROCEDURE 1

1. Wire the circuit shown in the schematic diagram of Figure 12.1.
2. Set your oscilloscope for the following approximate settings:
   Channels 1 and 2: 0.2 V/division, ac coupling
   Time base: 1 ms/division
3. Apply power to the breadboard, and adjust the input signal voltage to 1V peak-to-peak at a frequency of 100 Hz. ***You should make this voltage setting as accurate as possible.***
4. With the resistor and capacitor values used in this circuit, what do you expect the cutoff frequency to be? From the formula for cutoff frequency, what is the cutoff frequency?
5. With the input frequency set at 100 Hz, what is the peak-to- peak output voltage? What also do you observe?
6. Now vary the generator's frequency (fin), keeping the input voltage constant at 1 V peak-to-peak in order to complete the required data in Table 12-1. At the higher frequencies, you may have to increase the input voltage to obtain a measurable output level. Using the dB frequency response formula (Equation 5), calculate the expected dB response using a cutoff frequency of 710 Hz. Then plot both your experimental and your expected results on the blank graph provided for this purpose.

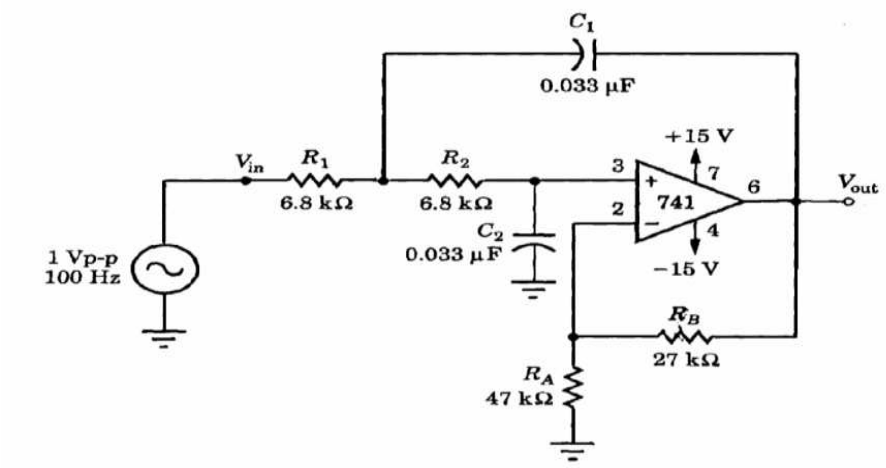
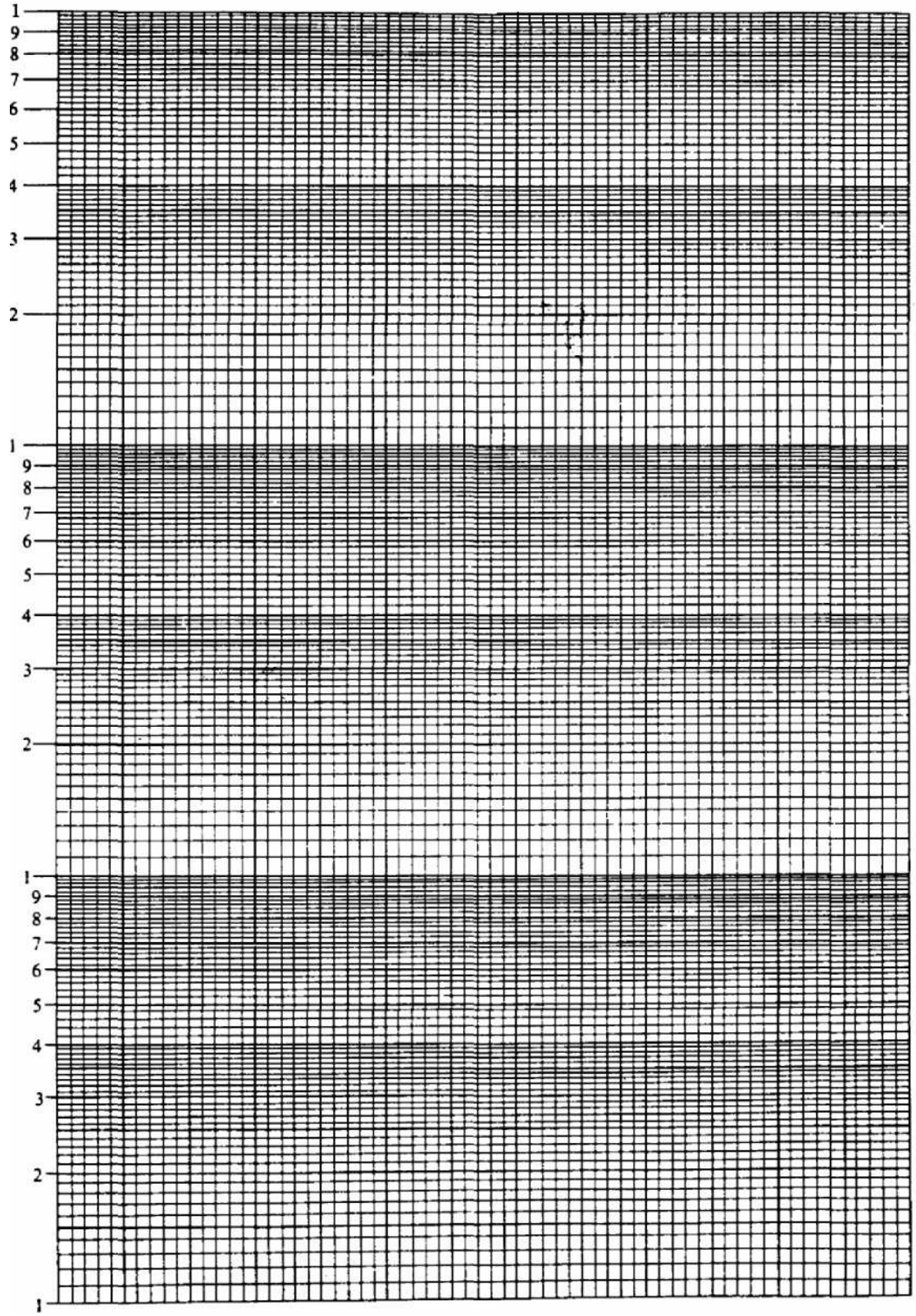


Figure 12.1

7. From your plotted results, you should find that both are parallel. How closely they are parallel will depend on how close the actual resistor and capacitor values are to the values shown in the schematic diagram.
8. Notice that the filter's gain at low frequencies is essentially constant up to some point (that is, the passband), after which it decreases at a linear rate with increasing input frequency. This linear decrease in gain as a function of frequency is termed the roll-off. To determine the roll-off of your filter, you must determine the slope of a line. Using the data in Table 9 -1, subtract the decibel gain measured at 1 kHz from that measured at 10 kHz. The frequency difference from 1 kHz to 10 kHz is one decade (that is, a factor of 10). Consequently, the roll-off is the difference in the decibel gain over a one-decade frequency range. From your measurements, what is the filter's roll-off and how does it compare with what you should expect for a Butterworth 2nd-order low-pass filter?
9. You should find that the low-pass filter's roll-off is nearly -40 dB/decade, or -12 dB/octave.
10. The filter's cutoff frequency is the frequency at which the dB frequency response is 3 dB less than the dB passband gain. This value is equivalent to an output voltage that is 0.707 times the input voltage of the filter. From your graph, estimate the filter's cutoff frequency, and compare it with the value calculated in Step 4. You should have estimated a critical frequency of approximately 710 Hz. In this case, the passband gain is 1.586, or 4 dB, so the critical frequency occurs when the filter's dB response is (4 dB - 3 dB), or +1 dB.

Input Frequency (Hz)	V_{in}	V_{out}	V_{out}/V_{in}	Experimental dB Gain	Expected dB Gain*
100	1 V				
200					
300					
400					
500					
600					
800					
1000					
2000					
4000					
5000					
6000					
8000					
10,000					

*Using a cutoff frequency of 710 kHz, for simplicity.


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